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SURVEY

Systematic Analysis of Federated Learning Approaches for Intrusion Detection in the Internet of Things Environment

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ABSTRACT The Internet of Things (IoT) has transformed various sectors by connecting devices, services, and applications, which boosts intelligence and operational efficiency. However, this growing connectivity has heightened concerns regarding data privacy and security, particularly for sensitive information. Intrusion detection systems (IDSs) have emerged as an effective strategy by incorporating machine learning (ML) methodologies to identify sophisticated cybersecurity threats. Traditional IDS solutions typically rely on centralized learning architectures, where data from edge nodes are transferred to central servers for analysis. This approach is effective but introduces major risks to the privacy of IoT data. Artificial intelligence (AI), specifically federated learning (FL), is a new frontier that offers an alternative option by enabling the direct implementation of distributed ML on IoT devices without requiring the sharing of raw data. FL supports decentralized and privacy-preserving learning, which makes it a critical instrument for enhancing the security of IoT ecosystems. Data security and privacy research in FL is increasingly becoming popular. However, the development of FL for IoT remains in its infancy and requires exploration in various areas. Moreover, current survey studies are limited and overlook the vital aspects of FL, particularly the diversity of datasets and the most recent methodologies employed in FL-enabled IDSs. This study conducts a systematic literature review that examines 43 studies published between 2020 and 2024. The focus revolves around recent advancements in applying AI to intrusion detection within IoT environments. This study comprehensively analyzes key ML models and datasets used to develop an IDS tailored for IoT systems. The review also explores emerging trends, challenges, and opportunities in implementing FL-based IDSs in IoT networks. The findings highlight the potential of FL to tackle pressing privacy and security challenges while enabling scalable and adaptive threat detection solutions. The study highlights important research gaps and proposes critical future directions to advance the field.

INDEX TERMS Intrusion detection system (IDS), Internet of Things (IoT), deep learning (DL), federated learning (FL), machine learning (ML), DDoS attacks.

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I. INTRODUCTION

The Internet of Things (IoT) represents the advent of a new era; it has revolutionized numerous sectors by integrating physical devices into interconnected networks, facilitating seamless connectivity and operation across different platforms. This integration has been pivotal in enhancing operational efficiency and opening new avenues for innovation and automation in healthcare, manufacturing, and consumer electronics. As devices become more interconnected than before, the ability of systems to operate harmoniously and exchange information effortlessly has represented a significant breakthrough in technological advancement [1], [2]. However, the rapid growth and complexity of IoT networks have introduced substantial security challenges. These issues can disrupt network functionality, leading to service disruptions that can hinder businesses, interrupt user services, and compromise critical infrastructure [3], [4]. The vulnerability of IoT systems to such attacks is specifically concerning given the often minimal security mechanisms implemented in connected devices and their inherent resource constraints; these restraints include limited computational power, memory, and bandwidth, which can leave entire networks susceptible to exploitation [5], [6]. Protecting these networks from attacks involves more than improving the security of individual IoT devices [7]. An extensive approach is needed to guarantee network security, such as rapid threat response, real-time monitoring, and continuous enhancement of defense mechanisms to counter evolving attack methods [8], [9].

Intrusion detection systems (IDSs) play a crucial role in strengthening the security of IoT. These systems are essential in providing a further line of defense capable of detecting suspicious activities, abnormal behavior, and unauthorized access in real-time, which mitigates potential threats before they can cause serious harm [10]. However, the unique behaviors and network traffic patterns should be monitored depending on the device or network segment type. This way allows for the provision of tailored security responses. Critical sectors such as smart homes, healthcare, and manufacturing rely on this flexibility to safeguard essential services against incursions that could compromise IoT systems, which place human lives at risk or interrupt vital services [11]. Centralized machine learning (ML) is often deemed unsuitable for IoT applications due to its prohibitive computational costs, inherent privacy, scalability, and resource-constrained vulnerabilities. Blockchain technology is a decentralized network operation that eliminates reliance on centralized authorities and enhances data privacy [12]. However, it still encounters great challenges [13].

Furthermore, modern approaches to IDSs in IoT networks have shifted toward decentralized models, such as federated learning (FL); FL is a decentralized ML paradigm in which several devices or entities can train a model simultaneously without sharing their raw data [14]. FL enables IDSs to learn from distributed datasets while safeguarding the privacy of individual devices [15], [16]. In addition to privacy preservation, FL offers enhanced adaptability and scalability in IoT networks. The data stays on local devices, which enhances resilience to cyber-attacks that target centralized systems and reduces the possibility of transmission data breaches [17]. As IoT devices continue to increase, the ability to scale IDS solutions while maintaining high levels of privacy and security becomes increasingly critical. With its decentralized architecture, FL delivers an effective solution to these challenges, specifically in environments where real-time threat detection and response are essential [18].

FL can be divided into three main forms: first is the horizontal FL, where the feature space of all clients is the same, but they differ in terms of data samples; second is the vertical FL, where the datasets across devices may have common samples, but the features are different; third and last is the federated transfer learning, which is applied when minimal overlap exists in features and samples between datasets [19], [20], [21]. FL facilitates highly robust and accurate detection of cyber threats, such as Distributed Denial of Service (DDoS) attacks [22]. With FL, the IDS may draw on the expertise of all participating nodes, which improves its capacity to manage the scale and diversity of IoT environments. Figure 1 shows a fundamental FL–IoT architecture.

A. MOTIVATION

Although FL has many advantages, implementing it in IoT security presents challenges. The heterogeneous nature of IoT devices, which range from low-power sensors to high-performance systems, introduces variability in computational capabilities and network connectivity [23]. This variability can affect the performance of federated models. Ensuring that FL models are sufficiently strong to withstand attacks is still a major concern. Additional studies need to be conducted to solve these issues, and more advanced ways should be developed to ensure the efficient operation and safety of FL-based IDS solutions when applied in various IoT settings [24]. Comprehensive security solutions are considerably and increasingly required, especially those that handle and protect the particular constraints of IoT devices while accommodating the restricted computational and energy capabilities. Some recent studies have investigated the deployed IDS to recognize threats in IoT using ML approaches. However, most of these studies cover certain aspects of this research field, while some require further investigation to present a thorough analysis of these works and methodologies. Furthermore, the literature review might be lacking from other published studies. Consequently, to our knowledge, this systematic literature review (SLR) study is considered the first to analyze research articles focusing on IDSs applying FL approaches in IoT systems.

B. CONTRIBUTION

The main contributions of this study are highlighted as follows:

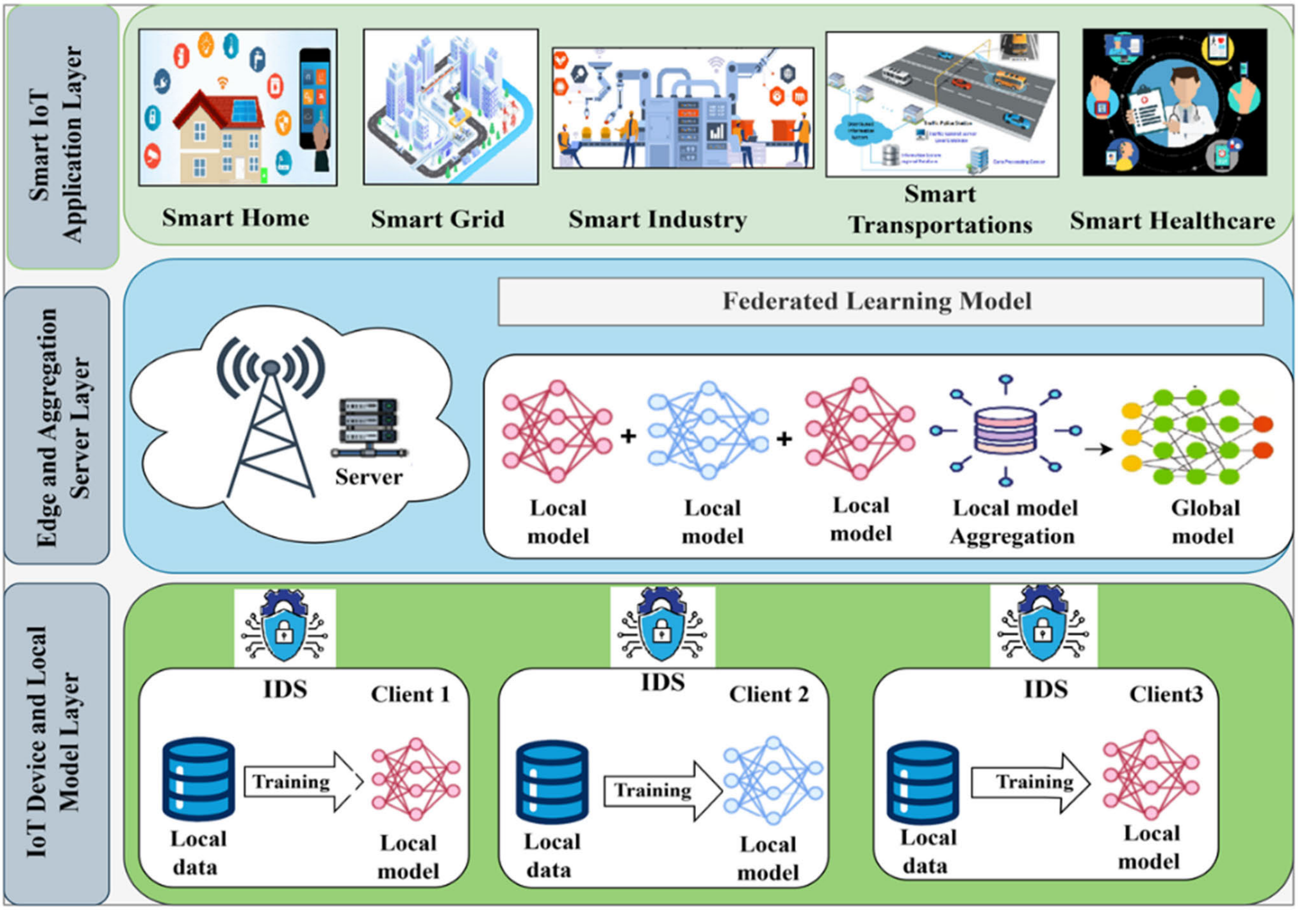

FIGURE 1. Architectural federated intrusion detection for IoT.

1) A comprehensive analysis of research articles directed on IDSs utilizing FL in IoT environments is conducted.

2) An inclusive taxonomy of ML models employed in FL-enabled IDSs is proposed using 43 selected studies. The taxonomy encompasses supervised, semi-supervised, unsupervised, and reinforcement learning (RL) approaches. Each category is examined in detail, with a critical assessment of its role and effectiveness within FL-based IDS frameworks.

3) Existing datasets are investigated and classified into general and specific IoT datasets for assessing FL-enabled IDSs with various performance metrics to evaluate the IDS.

4) The key challenges, limitations, research gaps, and future directions in the FL–IoT architecture are identified and discussed.

The rest of the paper is structured as follows. Section II reviews the related works. Section III provides our methodology, including the data selection criteria and procedures. Section IV addresses the research questions and presents a discussion of the findings. Section V presents the future direction. Section VI presents the limitations. Section VII elaborates the conclusions of this study. The overall structure and organization of this study are shown in Figure 2.

II. RELATED WORKS

FL has gained considerable interest recently as an alternative to traditional centralized ML methods [25]. Studies and surveys that provide an overview of FL have also substantially increased. However, as a new concept, FL still requires comprehensive studies and survey reviews on the IDS aspects of IoT. Table 1 highlights the list of the most current SLR, review, and survey studies, which explains their relevance to the works in this area.

Belenguer et al. [26] focused on the numerous applications of FL in IDSs based on IoT. They provided comprehensive insights into FL and IDS technologies and reviewed the state-of-the-art methods. In addition, the authors proposed a strategy for further research by discussing the drawbacks of employing FL for IDSs in IoT, including data quality, computational resources, privacy, and security issues.

To increase the efficiency of IDS for IoT, the authors in [15] surveyed and compared several studies that employed FL and addressed the primary challenges associated with this research. Data partitioning modeling among clients was the main focus of the writers. However, the researchers did not comprehensively analyze features like datasets, execution library, or ML technology. Their effect on potential IDSs in FL was also ignored. The scope of the review is limited, given that it referred only to 11 works.

The authors in [16] reviewed decentralized FL for IDSs in IoT systems. The researchers focused on privacy-preserving IDSs utilizing ML and distributed ledgers, pinpointed existing challenges, and proposed future research avenues for IoT security. Furthermore, the authors addressed the constraints of centralized IDSs.

The authors in [27] comprehensively analyzed the application of FL in IDSs. The writers covered a range of IDSs, relevant ML methods, and associated problems. In addition, the authors focused on the difficulties in FL implementations. This study provides a sense of potential future research avenues and establishes a baseline for subsequent works.

The researchers in [28] systematically delved into the modeling process of Fed-AIDS utilizing anomaly-based IDSs for IoT devices. They compared it with other works that frequently concentrated on analyzing security attacks on IoT devices. This method provides an improved and safer option. The writers addressed taxonomies that include model validation metrics, training datasets, technological complexity, classifier roles, and aggregation activities. The study aims to promote IoT device security by emphasizing several obstacles and suggesting plausible solutions to serve as a foundation for future studies on the Fed-AIDS model.

Lavaur et al. [29] surveyed FL approaches for IDSs. A taxonomy to assess the present state-of-the-art methods was also established according to the extent of each system (partial or complete) concerning security, confidence, and related factors. The authors studied the application of FL-based IDSs across multiple domains and emphasized the differences across various structural configurations.

The authors in [30] presented an SLR technique for evaluating recent publications in the FL-based IoT sector. The researchers examined 39 publications between 2018 and March 2022. Furthermore, the authors assessed the assessment factors, which include a 29% accuracy factor in the FL-based IoT field, a 23% epoch, an 18% time, a 9% energy consumption, a 9% delay, a 6% communication overhead, and a 6% privacy component. Finally, they reviewed future research challenges and open areas related to FL-based IoT.

The authors of [31] presented an overview of the integration of FL, IoT, and WSNs. Their work explores the integration of FL across various domains and delves into its fundamentals, techniques, and diverse applications. The authors also summarized current studies on FL heterogeneity and discussed associated issues. Methods for evaluating performance and concerns regarding security and privacy were also covered. In light of recent technical developments, the researchers highlighted the importance of the surveyed subjects and provided an overview of recent successes and prospects for research in FL, IoT, and WSNs.

The authors of [32] presented an in-depth review of FL security and privacy of data in the Internet of Medical Things (IoMT) networks. Specifically, they thoroughly assessed leading solutions to privacy and security problems with the Internet of Healthcare Things (IoHT). Furthermore, the writers classified the most pertinent articles regarding data security in the IoHT, highlighting developments in FL architecture or data protection. The writers covered various datasets from IoHT networks for training models. Data privacy and security in IoHT networks employing FL present many obstacles; however, this study aims to guide and inspire future research.

The primary characteristics of FL in the IoT were outlined by Yaacoub et al. [33] from the perspectives of privacy and security. The authors researched advanced FL algorithms, models, and protocols in terms of their effectiveness and practical applicability within IoT networks and systems. Data analysis and learning training negatively impact the current variety of devices and models within complicated IoT networks, which the researchers highlighted in their work. In addition, the proposed work offers a set of recommendations and suggestions that can boost the effectiveness of FL adoption and achieve higher attack resilience, particularly in dynamic IoT networks with heterogeneous components and devices with limited resources.

A comprehensive discussion of FL threats is presented in [34]. Rodriguez et al. provided an overview of threats in terms of a taxonomy of attacks and defense methods. The authors provided a wide range of possible client and server-side vulnerabilities. The surveys comprehensively examine security concerns in FL by investigating various security attacks according to their objectives, characteristics, and impact on the FL ecosystem. The survey examines threats across all three categories: horizontal FL, vertical FL, and federated TL. The authors argued that the dispersed nature of FL renders it susceptible to several hostile attacks. Consequently, they limited themselves to research on assaults on the federated model for altering its functionality and privacy breaches. In this way, sensitive information can be extracted from the learning process.

Gugueothet et al. [35] reviewed the application of FL and deep learning (DL) algorithms for IoT security. In contrast to traditional ML methods, FL models can preserve data privacy while exchanging information with other systems. Consequently, the authors proposed that FL can mitigate the limitations of traditional ML methods regarding data privacy during information exchange with other systems. The article examines several models, offers an overview, and compares and summarizes FL- and DL-based strategies for IoT security.

Nguyen et al. [36] comprehensively surveyed the increasing use of FL applications in IoT networks. The writers discussed the most recent developments in FL and IoT. They specifically examined the potential of FL for various IoT services, including mobile crowdsensing, attack detection, localization, data sharing, data offloading and caching, and IoT security and

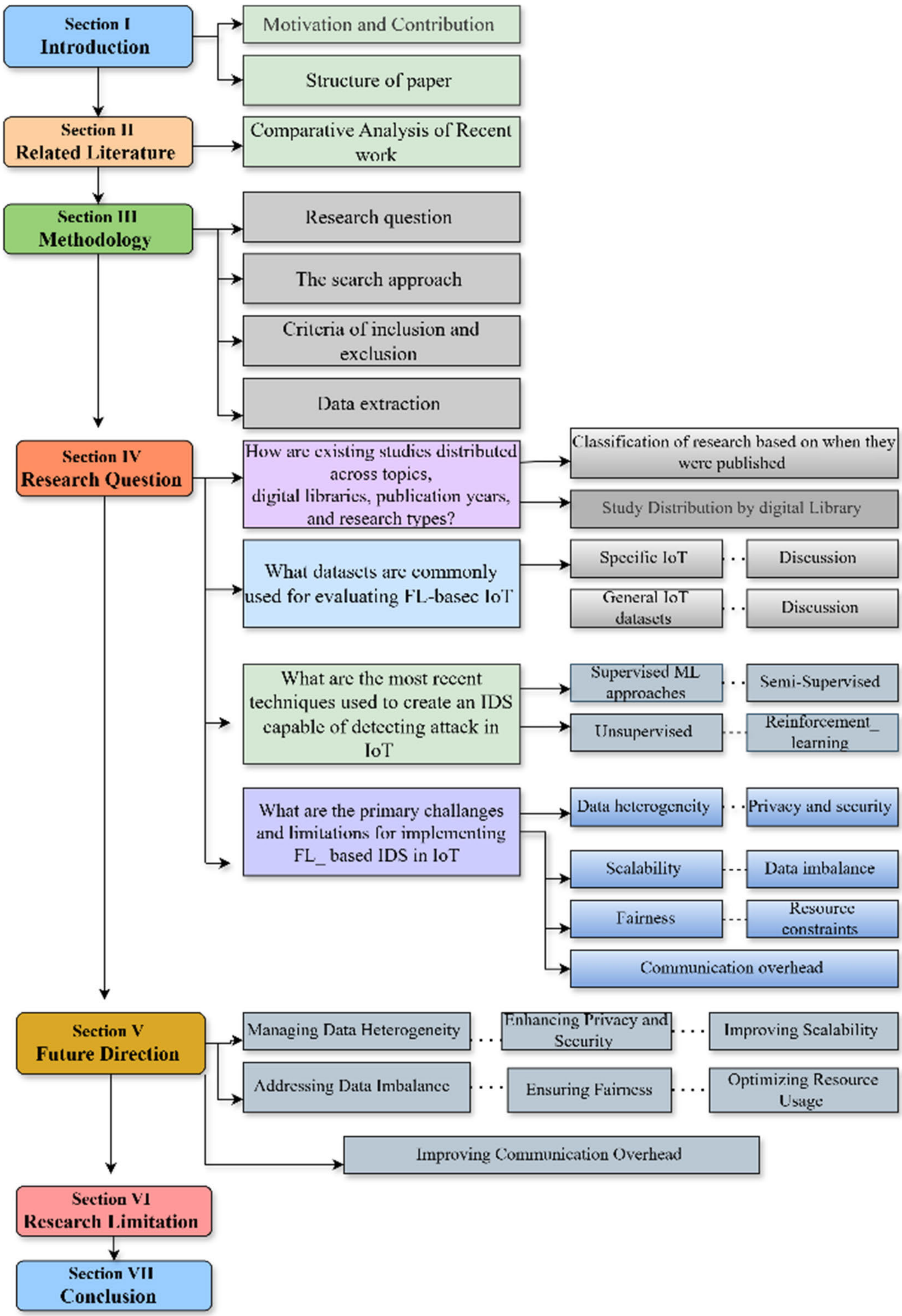

FIGURE 2. Organization of this article.

privacy. The authors further assessed the use of FL in different critical IoT applications, such as intelligent industrial, smart cities, smart transportation, smart health-care, and uncrewed aerial vehicles. Finally, this survey focuses on the existing difficulties and promising avenues for future research in this thriving field.

A systematic and transparent review was performed by Neto et al. [37] to identify relevant research for analysis. The researchers presented a framework for secure FL solutions in real-world applications and performed a security analysis of FL applications and ideas. This study additionally defines the security weaknesses of FL concerning model performance and data privacy breaches. The authors analyzed and compared possible mitigation solutions for these assaults.

Table 1 provides a comparison between existing literature reviews and the proposed SLR. The comparison reveals that no prior SLR has comprehensively addressed detection techniques, dataset analysis, and their application within the IDS domain of FL in IoT environments. Furthermore, only two studies, those by Belenguer et al. [26] and Umar et al. [28], partially cover these aspects.

The table below reveals that an SLR study is needed to analyze the current literature on FL-based IDSs for IoT devices. As a result, we will conduct a critical analysis of the available literature and research, relevant datasets, challenges, and some potential solutions. Our findings could help academics conduct further research to improve FL-based IDS techniques for detecting attacks in IoT networks.

III. METHODOLOGY FOR SYSTEMATIC REVIEW

An SLR intends to discover, analyze, and synthesize all the existing works in a specific area of interest. This type of study must be completed using a comprehensive, unbiased, and fair search strategy. This section summarizes the methodological steps of our SLR research, which are based on the PRISMA guidelines—Recommended Reporting Items for Systematic Reviews and Meta-Analyses. The significant steps of the procedure involve searching for related studies, filtering information from gathered papers, extracting data from subjectively selected studies, describing results, and proposing technical and practical ways in a theoretical approach to SLR studies.

A. RESEARCH QUESTIONS

This research addresses four key questions to explore the development and adoption of FL-based IDSs in IoT environments. The four research questions (abbreviated as RQ in this part) are provided as follows:

A. *RQ1: ow are existing studies distributed across topics, digital libraries, publication years, and research types?*
B *RQ2: What datasets are frequently utilized for assessing FL-based IDSs in IoT systems?*
C *RQ3: What recent techniques have been used to develop an IDS capable of detecting attacks on IoT devices?*
D. *RQ4: What are the primary challenges in implementing FL-based IDS approaches in IoT?*

B. SEARCH APPROACHES

This study focuses on IDS approaches for IoT devices utilizing FL methodologies. The SLR analyzes studies published between 2020 and 2024. Using targeted keywords derived from the research questions, relevant scholarly articles were identified through automated searches conducted across key academic databases, including Google Scholar, IEEE Xplore, ScienceDirect, Springer, and Web of Science. Figure 3 shows the PRISMA flow diagram [38], providing a structured overview of the procedures related to the article selection process, which provides a structured overview of the procedures related to the article selection process.

We conducted a literature survey in three stages: collecting information, filtering articles, and reviewing them. The research keywords "federated learning" AND "Internet of Things" OR "IoT" AND "Intrusion Detection" were used to identify suitable search terms for the issue at hand. Therefore, to narrow the research scope of the targeted topics and subjects, more restricted and targeted search terms were utilized based on the articles identified and reviewed using keywords with ("federated learning") AND ("Internet of Things" OR "IoT") AND ("Intrusion Detection System" OR "IDS") AND ("Distributed Denial of Service" OR "DDoS"). Research papers were extracted from the identified databases. To determine whether the retrieved articles fulfilled the inclusion or exclusion criteria, the second phase involved filtering them based on their titles and abstracts. The third phase explored the content of each research.

C. CRITERIA FOR INCLUSION AND EXCLUSION

This subsection outlines the inclusion and exclusion criteria applied in the SLR. First, only scholarly works published between 2020, and March 2024 were solely examined (to ensure relevance). Second, the SLR included only publications in English to guarantee accessibility and understanding. Third, only journal articles and conference papers were analyzed, while other categories, such as books, reports, editorials, reference works, and dissertations, were excluded.

The "kind of documents" filter was used in most databases to facilitate this exclusion, except for Google Scholar, where retrieved studies were manually sorted by type using EndNote software. Fourth, duplicate articles identified across databases were removed using EndNote. Finally, abstracts were screened to finalize the selection of studies, focusing on FL-enabled IDS approaches incorporating ML and DL in IoT networks.

Figure 3 illustrates the total number of investigations at each methodology step. The first stage involved retrieving 226 studies from four digital databases (i.e., Google Scholar, ScienceDirect, IEEE Xplore, and Web of Science) by using keyword searches.

During the screening process, the primary author initially selected titles and abstracts. The second reviewer was consulted in cases of ambiguity or uncertainty regarding the inclusion or exclusion criteria for specific studies.

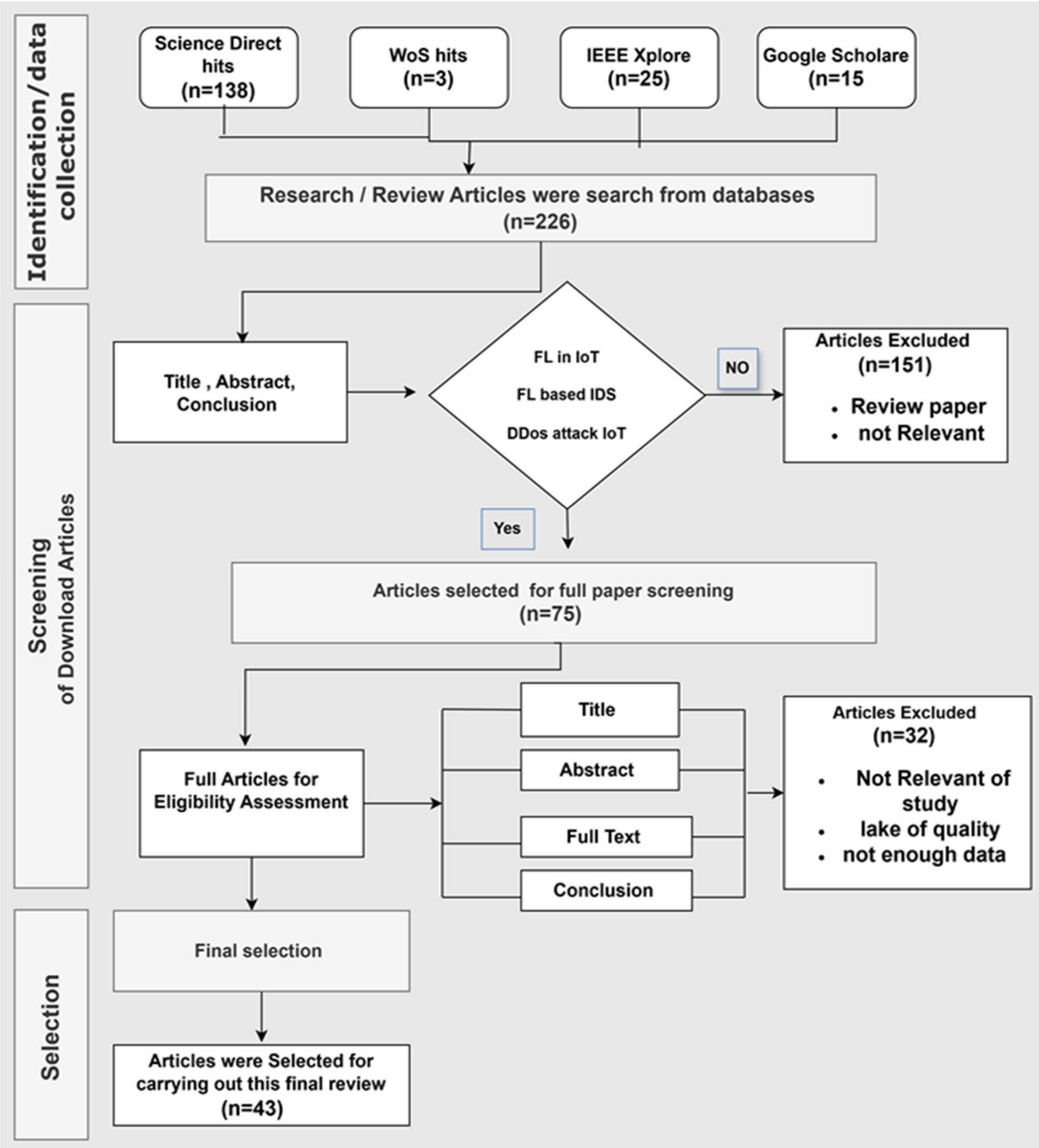

FIGURE 3. The PRISMA flow diagram for the selection of articles comprises identification, screening, and selection steps.

However, all discrepancies were discussed until a consensus was achieved.

For the initial screening, 151 studies were excluded from the title, abstract, and keywords of each study. The screening process identified 75 papers. The screening was conducted on the previously selected studies to choose the qualifying studies. Next, 32 articles were eliminated from the reports evaluated for eligibility due to exclusion criteria in the full text of the publication. Finally, only 43 articles were included in the SLR study. Table 2 summarizes the filtering of literature based on these criteria, while the PRISMA flow diagram illustrates the overall methodology.

D. DATA EXTRACTION

This step focuses on the data extraction procedure, which involves extracting information from each paper and synthesizing it in two steps: (a) extracting data and (b) evaluating the quality of the article [39]. These processes were conducted on

TABLE 1. Comparative analysis of recent literature review.

References	Year	Main Finding	FL	IDS	IoT	IDS/IoT Dataset	Detection Techniques	challenge	Taxonomy	Research Direction
[Assis Moreir, et al][16]	[2021]	Discusses IDS methods in decentralized, federated learning and distributed strategies for IoT security, highlights challenges for IDS in IoT	✓	✓	✓	△	×	✓	×	✓
[Agrawal et al.] [27]	[2021]	Overview of federated learning implementations in anomaly detection, highlighting issues and potential research directions FL-based IDS.	✓	✓	×	×	△	✓	×	✓
[Rodriguez, et al] [34]	[2021]	Taxonomy of the threats of federated learning and defence methods	✓	×	×	×	×	✓	✓	×
[Nguyen et al.] [36]	[2021]	Cover FL applications in IoT, discuss recent advances, integration potential data sharing, attack detection, and privacy, highlight challenges, and suggests future research directions.	✓	×	✓	×	△	✓	✓	✓
[Belenguer,, et al] [26]	[2022]	Explores FL applications in IDS, discusses both technologies, categorizes current research, identifies limitations of FL/IDS	✓	✓	✓	△	△	✓	✓	✓
[Novikova, et al] [15]	[2022]	Highlights the advantages, challenges, architectures, and data partitioning approaches of FL-based AIDS.	✓	✓	✓	△	×	✓	×	×
[Umar et al.] [28]	[2024]	Explain a technical analysis of Fed-AIDS modeling for IoT security	✓	✓	✓	△	△	✓	×	✓
[Lavaur et al.] [29]	[2022]	Covers FL-based IDS from 2016 to 2021, taxonomy for designing and comparing FL-based IDS, Identifies research directions.	✓	✓	×	△	✓	✓	✓	△
[Hosseinzadeh] [30]	[2022]	Taxonomy of FL-based IoT impact, evaluates factors, Highlights difficulties, and future research directions in the context of FL-based IoT.	✓	×	✓	△	×	✓	✓	✓
Yaacoub et al. [33]	[2023]	Explain FL security and privacy features in IoT, analyze advanced FL algorithms, models, and protocols, and compares cryptographic and non-cryptographic protection solutions.	✓	△	✓	×	×	✓	✓	✓
[Gugueoth et al.] [35]	[2023]	Analyzes various ML algorithms for IoT security, and highlights advantages and limitations, with a focus on implementing FL and DL for enhanced IoT security.	✓	✓	✓	×	✓	×	✓	×
[Neto, et al] [37]	[2023]	Explore Applications, Attacks, and Challenges of FL	✓	△	×	×	×	✓	×	×
[Mengistu, et al] [31]	[2024]	Discusses the integration of FL with IoT and WSNs, covering FL fundamentals, strategies, challenges of heterogeneity, security, privacy, and performance evaluation methodologies.	✓	×	✓	✓	×	×	✓	✓
[Coelho et al.] [32]	[2024]	Explores FL's data security and privacy applications in IoHT networks, covering key concepts, categorizing relevant studies, and highlighting challenges and future research directions in IoHT networks.	✓	×	✓	×	×	✓	×	✓
Our Work	[2024]	The above research lacks a comprehensive investigation of detection techniques and dataset analysis, in IDS domains of FL in IoT.	✓	✓	✓	✓	✓	✓	✓	✓

The symbol "✓" indicates comprehensive coverage of concepts, "△" signifies partial coverage, and "X" denotes that the scope is not cover all 43 articles and selected literature, with Endnote Reference Manager abstracting all relevant data. A table was created to summarize all relevant data extracted from the final selected studies—the process aimed to organize key information and assess the quality of the evaluated articles. The extraction form included article identification (title, authors, source, year) and specific methodological and outcome-related data. Table 3 presents the strategy for retrieving data from the 43 studies.

TABLE 2. Criteria for inclusion and exclusion in the relevant literature search (Article Filter).4.

Inclusion Criteria	Exclusion Criteria
• Answered research questions.	• It did not directly address the question raised by the study.
• Studies published in the years 2020 to April 2024.	• Articles published before 2020.
• Only academic papers	• Non-academic publications
• Studies focused on FL-based IDS in IoT in the context of security and privacy	• Studies not focused on FL-based IDS in IoT
• We selected articles relevant to DDoS and IoT, DDoS and IDS, DDoS and FL, FL and IDS, FL and IoT.	• Avoid articles unrelated to IoT DDoS, FL, and IDS, FL, and IoT
• Only papers written in the English language.	• Papers that were not written in English.

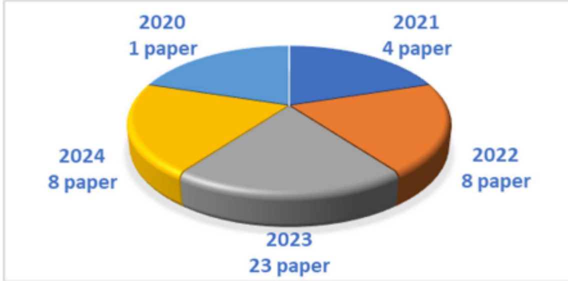

FIGURE 4. Analysis of the selected research by publication year.

TABLE 3. The data extraction forms the information selected from reviewed article.

Data extracted	Description
Bibliographic information	• Title of publication Journal • Author(s). • Journal domain. • Year of publication
Publication Types	• Journal Article • conference paper
Methodology	• Objective of the study • Research methods (Classification and Detection mechanism, different algorithms, machine learning models, deep learning models used, etc.). • Data sources (the public or private datasets utilized). • Detected attacks using proposed IDS models in determined papers.
Studied focuses	• A model or set of factors for identifying attacks. • Methods • features
Results	• Study Finding • limitations of the proposed FL-based IDS model • Conclusions

IV. RESEARCH QUESTION

A. RQ1: HOW ARE THE IDENTIFIED STUDIES DISTRIBUTED ACCORDING TO SUBJECT, DIGITAL LIBRARY, PUBLICATION YEAR, AND PUBLISHING TYPE?

1) CLASSIFICATION OF RESEARCH BASED ON PUBLICATION YEAR

The distribution of selected research publications from 2020 to 2024 is presented in Figure 4. The data reveal a considerable increase in research published over the years, with the majority published in 2023, amounting to 23 studies. This trend indicates a growing interest in the topic, particularly in the last decade. The few works conducted before 2023, such as only one in 2020 and four in 2021, suggest that the field was relatively nascent or under-researched during those years.

However, the substantial increase in 2023 signifies an acceleration of interest due to technological progress, methodologies, or a shift in academic and industrial priorities regarding the research topic. The observed growth in publication numbers in 2023 is an important finding, which highlights the emergence of new research directions and potentially improves accessibility to data or resources that have facilitated such an increase. On the contrary, the slight drop in publications in 2024 (7 studies) may indicate a stabilization of research outputs or a potential consolidation phase where researchers focus more on validating or implementing existing findings rather than exploring entirely new avenues.

V. STUDY DISTRIBUTION BY DIGITAL LIBRARY

The distribution of studies across various digital libraries, which include IEEE, ScienceDirect, Google Scholar, Springer, and Web of Science, is illustrated in Figure 5. The most significant proportion of selected studies comes from ScienceDirect (21 studies), followed by IEEE (15 studies).

This distribution demonstrates the dominance of these two digital libraries in disseminating research on the topic. Additionally, these platforms serve as primary sources of credible and impactful research in the field, thereby emphasizing their role in curating high-quality, peer-reviewed articles. On the other hand, Google Scholar, which accounted for 4 studies, and Springer, with only 2 studies, represent a smaller fraction of the selected literature. Notably, no studies were found in the Web of Science, indicating either the limited relevance of this database to the topic or possible exclusion criteria that impacted its representation. The limited presence of Springer and Google Scholar studies suggests that researchers in this domain might prefer repositories known for high-impact journals, such as ScienceDirect and IEEE, which likely provide better access to peer-reviewed and well-cited literature.

The trends observed in the distribution of selected studies by year and digital libraries suggest several implications for future research. The spike in publications in 2023 points to a concentrated body of recent work that can serve as a foundation for systematic analysis and synthesis. Researchers may focus on understanding the underlying themes and methodologies of these studies to identify existing gaps, particularly from earlier years where data is sparse. The limited number of studies from specific digital libraries, such as Springer, and the absence of studies from the Web of Science suggest an opportunity for future researchers to explore and incorporate less-utilized sources, which may contain valuable but overlooked information.

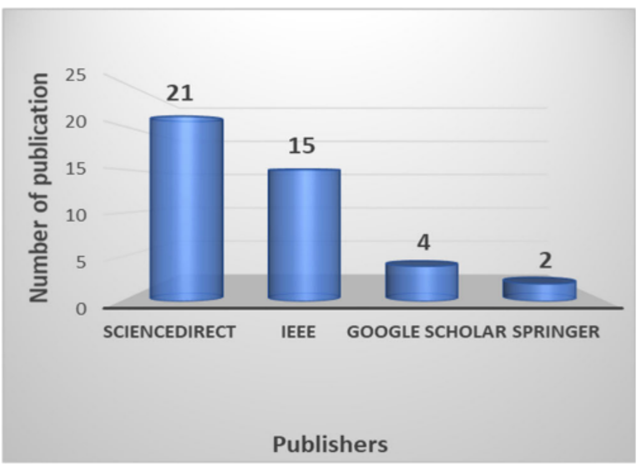

FIGURE 5. Distribution of the selected studies according to digital libraries.

The concentration of studies within specific digital libraries also raises questions about the accessibility and diversity of research in the area. A heavy reliance on a limited number of databases may create a bias towards certain studies. Therefore, future researchers should consider broadening their search to include multiple digital libraries, potentially diversifying the literature and contributing to a more complete grasp of the topic.

A. RQ2: WHAT DATASETS ARE COMMONLY USED FOR EVALUATING FL-BASED IDS IN IoT SYSTEMS?

This question aims to explore available datasets and analyze their properties to evaluate IDS-enabled FL models. Datasets represent an essential part of designing and developing effective IDS solutions for IoT devices. The majority of the reviewed work utilizes publicly accessible datasets due to the challenges associated with directly gathering both attack and normal IoT traffic. We categorized the dataset into IoT-specific situations and general internet datasets. The dataset is compared through many measurements, including data type, feature count, sample size, and attack style, as illustrated in Tables 4 and 5.

1) SPECIFIC IoT DATASETS FOR ASSESSING THE FL-ENABLED IDS DATASET

- The N-BaIoT [40] is a new network-based dataset that concentrates on botnet attacks on IoT networks. Specifically, it simulates two known botnets: Bashlite and Mirai [41]. These botnets can launch TCP and UDP flooding attacks, including port scanning. The authors evaluated the dataset using autoencoder models.
- The Bot-IoT dataset [42] is developed at the UNSW Canberra Cyber Range Lab by simulating a realistic network environment with heterogeneous network profiles. It includes labeled data for various attack types, such as DoS, scans, DDoS, and data-stealing assaults, performed on five devices. This dataset is critical for designing and evaluating intrusion detection systems (IDS) and enhancing IoT security. The dataset is offered in many formats, namely original Pcap files and CSV files. The Pcap files alone amount to 69.3 GB, comprising over 72 million records. This dataset focuses on IoT-specific protocols such as Message Queuing Telemetry Transport (MQTT). New characteristics were retrieved utilizing the Correlation Coefficient and Joint Entropy approaches. The extracted features are subsequently input into different ML and DL algorithms to determine detecting attack accuracy.
- Using the IoT-Flock traffic generator, the MQTTset dataset [43] captures MQTT protocol traffic from various IoT devices, including door locks, temperature sensors, motion sensors, and fan speed controllers. The dataset, provided in a CSV file, includes 33 features and data from various attacks like flooding and DoS. MQTT publishes flood and malformed data. The dataset is passed on to different machine learning algorithms, reporting accuracy.
- Edge-IIoTset dataset [44] offers a realistic resource for cybersecurity research in IoT and IIoT applications, facilitating ML-based IDS development in centralized and FL modes. It incorporates numerous IoT devices and edge computing nodes, capturing both normal and malicious traffic across various IP-based protocols. The application-layer protocols included in the dataset are HTTP, MQTT, DNS, and Modbus. It includes 14 attacks on IoT/IIoT protocols, such as denial of service, man-in-the-middle, malware, and injection.

Aiming to address existing datasets' limitations, it provides full packet captures in PCAP format. Following data collection, 61 traffic features, including attack labels, were extracted during preprocessing.

- The TON-IoT dataset [45] is gathered from a testbed that simulates a real-world smart city network, incorporating edge, fog, and cloud computing services. A heterogeneous dataset for IoT/IIoT based on telemetry and data-driven methodologies. The collected data includes both normal and assault samples. The dataset accurately simulates real-world network traffic and includes a comprehensive range of attack subcategories alongside seven IoT devices. The simulated devices consist of those featured in the Bot-IoT dataset, a Modbus industrial control device and a GPS tracker. 44 features were extracted into four categories: Connection features, User-centric features, Statistical features, and Violation features.
- The Kitsune dataset [46] is generated from traffic provided by an eight-IP camera video monitoring system comprising nine IoT devices, including two smart doorbells, a baby monitor, and a thermostat. The dataset encompasses four categories of cyberattacks, notably MiTM, reconnaissance, and DoS attacks. It includes over 100 extracted features relevant to network intrusion detection. Both CSV format and pcap versions of the dataset are available. Designed as a plug-and-play Network Intrusion Detection System to identify normal and malicious samples.
- Human Activity Recognition (HAR) [47] is publicly available and commonly used in ML and signal processing to develop models and to identify human activities. The HAR dataset is always collected by having people do predetermined activities while wearing devices outfitted with sensors. The dataset often comprises sensor readings like accelerometer and gyroscope data obtained from smartphones or other wearable devices. Every data segment is labeled based on the activity conducted during that interval.
- NF-ToN-IoT-v2 [48] is produced by processing the respective .pcap files of the ToN-IoT dataset with the standard NetFlow tool. It produces a database with 43 feature dimensions of values. It is designed to have each network flow record with full and accurate labeling for a wide range of attack types and thus is ready for machine learning-driven network intrusion detection systems.
- The Real Urban IoT dataset [49] is the key to mitigating DDoS attacks, yet the lack of IoT-specific datasets poses a significant challenge. The authors introduce a dataset from a real-world urban IoT testbed featuring 4,060 event-driven sensors distributed across a major city to address this. The dataset logs binary activity status for each node at 30-second intervals over one month and includes metadata fields for node ID, geolocation, and activity timestamps. Additionally, an accompanying script enables the simulation of attacks, with benign activity labeled "0" and attacks as "1." Both the dataset and the attack simulation script are provided in CSV format.
- The CIC-IoT22 [50] is developed using 81 IoT devices, including cameras, smart lamps, speakers, doorbells, motion sensors, and weather stations, which communicate through Wi-Fi, Zigbee, and Z-Wave protocols. Although the dataset primarily targets device identification and profiling, it also includes six experimental scenarios, including simulations of denial-of-service (DoS) and Real-Time Streaming Protocol (RTSP) brute-force attacks.
- Wireless Sensor Network Dataset (WSN-DS) [51] is explicitly designed for evaluating IDSs within wireless sensor networks (WSNs). This dataset provides valuable data for researchers working on security solutions for WSNs. These solutions are critical in various applications such as environmental monitoring, military surveillance, and healthcare. The dataset contains benign and various malicious activities, with four categories of DDoS attacks: Blackhole, Scheduling, Grayhole, and Flooding attacks. The entire dataset comprises 374,661 labeled records. The dataset also consists of 23 attributes, including the node ID, the number of transmitted and received packets, and the topological data of the WSN.

2) DISCUSSION ON THE SPECIFIC IoT DATASET

Choosing appropriate datasets is critical in assessing the performance of IDSs designed for IoT environments. This section reviews the datasets utilized in the literature to test the efficiency of proposed IDS models, with emphasis on their relevance and adequacy for addressing IoT-specific security challenges. IoT-specific datasets play a dual role: they are essential for the training and testing phases of IDS development. These datasets reflect the unique characteristics of IoT networks, including their constrained resources, diverse protocols, and distributed nature, which make them indispensable for accurate evaluation. Several significant points may be highlighted by examining the datasets. The most prevalent datasets are TON-IoT, N-BaIoT, BoT-IoT, Edge-IIoTset, and MQTTset. These datasets precisely capture IoT scenarios. They address the unique characteristics and attack vectors associated with IoT devices and networks, which differ from those in general network traffic. In contrast to research using conventional network traffic datasets, 30 studies employed IoT-specific datasets to highlight their importance in ensuring the accuracy and relevance of IDS models in IoT environments. Several conclusions can be drawn based on the previous description and the information provided in Table 4. First, the generation of IoT-related datasets has evidently increased in recent years, which suggests a growth in the implementation of IoT technologies. However, we found that around 12 proposed works use datasets before 2019, which do not sufficiently represent modern communication networks. Second, regarding the benign–malicious attack ratio, although a higher proportion of benign traffic characterizes real-life scenarios, we verified that a significant disparity exists between the different datasets. Specifically, some of them (e.g., Bot-IoT or IoT network intrusion datasets) contain a stark contrast between benign and malicious samples (9,543 vs. 73,360,900). Third, most datasets do not contain modern attacks, while a significant portion embraces a small number of attacks. According to our analysis, most datasets can be realistically divided (e.g., using the IP address) to be used in an FL scenario. However, datasets such as Kitsune, IoT-23, and MQTTset are not directly applicable. They must be previously preprocessed such that the network traffic from the same IP address can be considered a device acting as an FL client. In addition to the analysis of the described datasets, Table 4 shows which datasets are used by the suggested FL-enabled IDS in IoT. Figure 6 illustrates the various types and frequencies of datasets used in the selected studies.

3) GENERAL IoT DATASETS FOR ASSESSING THE FL-ENABLED IDS DATASET

Generic datasets are vital for IDS research, although they are not specifically tailored to IoT environments. Historically, datasets, such as KDD Cup 99 [52],are used extensively to evaluate IDS models in IoT before IoT-specific datasets became available. The relevance of non-IoT-specific datasets persists because many IoT devices use generic protocols such as TCP/IP and HTTP, which render them vulnerable to threats similar to those targeting general-purpose computing systems. Consequently, datasets capturing traffic associated with these protocols can still provide valuable insights for IoT-related research.

Figure 7 illustrates the types and frequencies of datasets utilized in the analyzed studies. Among these datasets, the most frequently used is CICDDoS2019, followed by NSL-KDD, CIC-IDS2017, UNSW-NB15, and CSE-CIC-IDS2018. Notably, these datasets predominantly focus on general internet contexts and are not explicitly designed for IoT-specific scenarios. The total number of studies conducted in generic internet contexts is 43. We will examine datasets focused mainly on general online contexts in this area. The following paragraph briefly discusses the most widely employed datasets in the analyzed research.

- UNSW-NB15 [53] is a publicly available resource that is widely used in research on network intrusion detection. It was generated using the IXIA PerfectStorm tool at the Cyber Range Lab of the Australian Center for Cyber Security, University of New South Wales. The dataset was generated by monitoring network traffic over 16 days, encompassing legitimate and malicious activities. The data collection simulates standard operations and a range of malicious assaults,

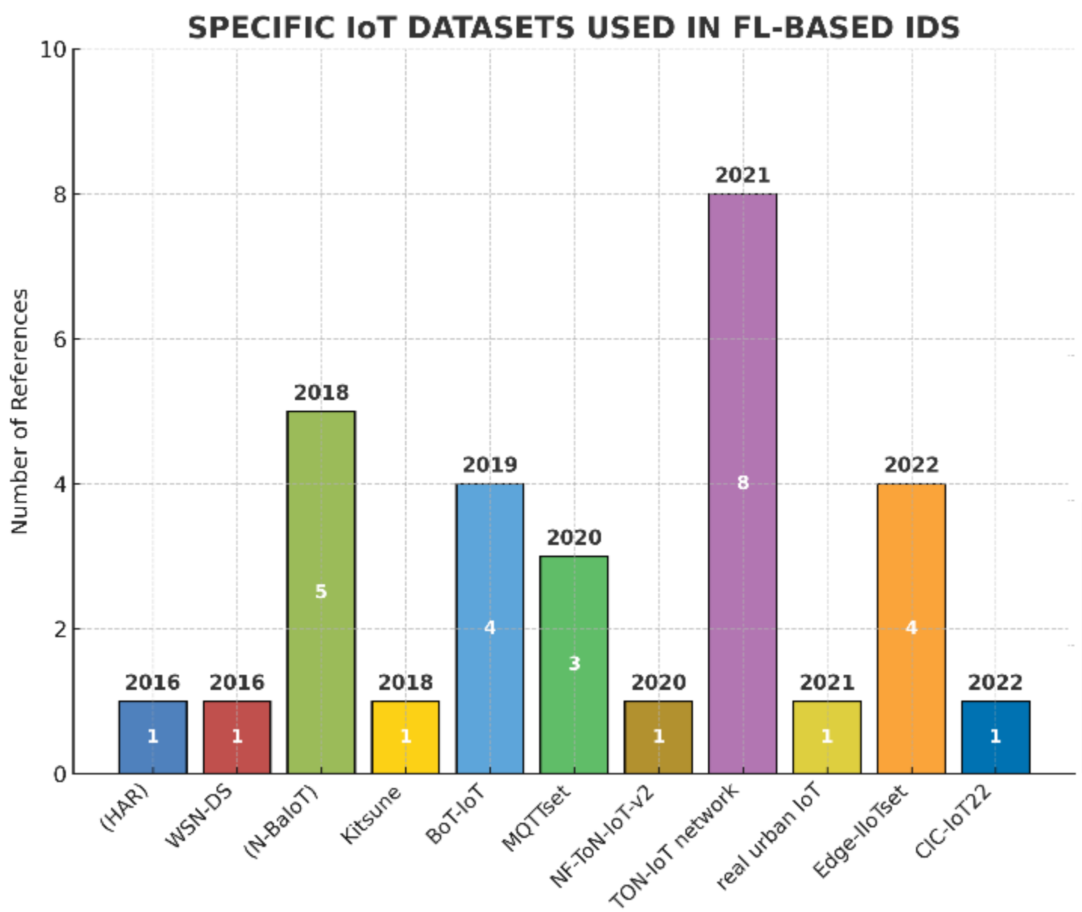

FIGURE 6. Published Studies are distributed according to the year of the IoT dataset.

including nine distinct attack types: Backdoors, Exploits, Denial of Service (DoS), Reconnaissance, Fuzzers, Analysis, Generic, Shellcode, and Worms. The dataset includes 49 attributes grouped into flow, connection, basic, content, time, general, and labeled categories, which allows a comprehensive foundation to evaluate the performance of IDSs.

- In 2018, the Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC) collaborated to provide the CSE-CIC-IDS2018 dataset [54]. The network of interest contained five different organizational-type departments and an extra room for servers. The benign packet portion was generated based on ''abstracted representations'' from users in terms of a sequence of actions involving applications (i.e., active use of a software-based service) occurring at events on the network. At the same time, attack scenarios were conducted by one or several commodity machines against the victim network. The original dataset extracted all 83 features using the CICFlowMeter-V3 tool. The dataset consists of 16,232,943 flows, where 13,484,708 (83.07%) are benign and 2,748,235 (16.93%) represent attack traffic.
- CIC-IDS2017 [54] is a widely utilized dataset for IDS purposes. It was created in an emulated networking environment and comprises five days of network traffic. This labeled dataset covers various attacks, including brute force, DoS, network infiltration, and botnets. It contains 80 features associated with the network traffic extracted using the CICFlowMeter tool and provides information such as timestamp, source, destination, port, and IP address [55]. Furthermore, it can be downloaded in either PCAP or CSV format.
- USTC-TFC2016USTC-TFC2016 [56], which is another popular dataset, is divided into two components. The first section includes 10 types of malware traffic gathered from public websites operating in a real network environment between 2011 and 2015 [57]. The entire size of this dataset is 3.71 GB, and it is stored in the PCAP format.
- InSecLab-IDS-2021dataset [58] is constructed to emulate DDoS attack scenarios and capture related network traffic. The environment consists of a victim's SDN network and an attacker's network built using a Mirai botnet. The DDoS attacks target the SDN network under three scenarios: SYN Flood, UDP Flood, and ACK Flood. CICFlowmeter-v39 processes captured traffic to extract 80 features, which enables flow classification into Benign or Attack types (UDP, SYN, or ACK) based on IP addresses and timestamps. The final dataset contains 4,091,376 records, with a total of 1.51 GB.
- NSL-KDD dataset [59] is developed to address the main drawbacks of the widely criticized KDD99 dataset. This dataset provides a more balanced and complete test bed by removing redundant records and restructuring its format. It consists of three parts: KDDTrain+, KDDTest+, and KDDTest-21. In particular, KDDTest-21 is a special subset of KDDTest with novel attack types not present in KDDTrain+ or KDDTest+. Thus, it enables robust testing of detection models. This dataset classifies network traffic into five classes, which are R2L, Normal, DoS, U2R, and Probe. NSL-KDD retains 41 features per record of traffic to maintain continuity in comparative research, which is similar to KDD99.
- CICDDoS2019 [60] is a robust benchmark containing benign traffic and the latest DDoS attack types, resembling real-world scenarios. These attacks are organized into 13 categories: DNS, MSSQL, NTP, LDAP, SSDP, NetBIOS, SNMP, UDP, UDP-Lag, TFTPDDoS, WebDDoS, SYN, and PortScan, with codes ranging from 0 to 13. Timestamp features are included, which makes the dataset suitable for temporal and sequence-based analysis.
- KDDCup99 [52] is a benchmark dataset for NIDS and comprises roughly 4.9 million connection records. Each record contains 41 attributes, which are classified as normal or specific attack types. The dataset includes many simulated invasions in a military network environment, which supports establishing and validating ML models for anomaly detection. Despite its widespread use, the dataset has faced criticisms due to its shortcomings such as redundancy and an artificial distribution of normal and attack records. These limitations may influence the generalizability of models trained on this dataset. Researchers often preprocess the data to solve these concerns and increase the performance of IDSs.
- InSDN dataset [61] is tailored for SDN-specific network attacks. It facilitates the evaluation of IDS efficacy across many attack categories inside diverse facets of the SDN framework. It comprises around 340,000 recordings, including various attacks. The dataset is accessible through all capabilities of SDN elements. Every record in attack scenarios comprises 76 attributes.
- The Modbus network dataset [62] is utilized for cybersecurity research and development for IDS within industrial control systems. The dataset typically contains network traffic data captured from a Modbus network. This dataset includes various types of packets that follow the Modbus protocol, which details the communications between primary and secondary devices in the network. The dataset includes normal operational traffic and traffic generated by known attacks. Industrial automation solutions address the interoperability issue by combining IoT devices with the Modbus protocol.
- MNIST [63] is short for the Modified National Institute of Standards and Technology dataset. This dataset is a popular choice in ML and computer vision. It is used as a baseline for training and testing image processing models and for benchmarking classification algorithms' performance. The dataset comprises 70,000 samples of handwritten digits (ranging from 0 to 9), with each image normalized to 28 $\times$ 28 pixels and stored in a one-dimensional format. The data are split into training and test sets in a 6:1 ratio, with 250 examples provided for each digit class.
- The CICEV2023 DDoS Attack Dataset [64] facilitates training and testing features for the detection classifiers of

DoS and DDoS attacks. The dataset is specifically tailored to evaluate electric vehicle charging systems (EVs). It is generated using a simulator that models several EVs, charging stations (CS), and a grid server within a charging infrastructure network. The dataset includes four distinct attack scenarios: (1) Correct EV ID, (2) Incorrect CS Time stamp, (3) Incorrect EV Timestamp, and (4) Incorrect EV ID. The authors implemented an EV authentication procedure within the simulator to generate the dataset. The authors designed an EV authentication process to create the dataset. For scenarios (1) and (2), real-time data were collected using Perf, which is a Linux tool that is widely used to measure software performance [65], [66].

4) DISCUSSION OF THE GENERAL IoT DATASET

Recent analysis data descriptions, such as those presented in Table 5, show a considerable increasing trend in creating general datasets. The fast development of IoT devices has led to the requirement for robust data management and analysis frameworks to leverage IoT advances. The IDS datasets analyzed in this study (12 out of 23) primarily focused on general internet environments.

The CICDDoS2019 dataset is a widely utilized within general internet environments, as shown in Figure 7. This dataset offers a thorough and up-to-date representation of prevalent DDoS attack patterns, which makes it a valuable resource for evaluating IDS models in non-IoT domains. Most studies have reported high-performance metrics, including detection accuracy and efficiency. For example, some studies achieved an accuracy of 98.37% (computational time: 3.917 ms, ROC AUC: 0.9890) and 99.59% (false positive rate (FPR): 0.042%, detection time: 0.062 ms) [67], [68]. In addition, one study [69] tested various models and reported F1-scores ranging from 0.9667 to 0.9899, along with improved true positive rates and better accuracy for non-independent and identically distributed (non-IID) datasets. Convergence time is reduced as well. The comparison indicates that utilizing this dataset results in greater accuracy, precision, recall, and F1-score values. Only one study [70] demonstrates a significant variance in rating measures across similar datasets.

As shown in Figure 7, several works also used the NSL-KDD dataset [59], which was created in 2009 as an improved version (after removing duplicate packets/flows) of the KDD-Cup99 [52] dataset, which, although deprecated, is still widely used. These datasets focus on attacks related to DoS, privilege escalation, and probing. The majority of research produced high-performance metric values such as [71], [72], [73], and [74]. Only one study [75] showed low accuracy of FL, which ranges between 76.84% and 77.79%.

Another widely used dataset is UNSW-NB15 [53], which was proposed in 2015. It contains 49 features to characterize network traffic considering 45 IP addresses, which can be used for FL purposes. Moreover, the authors in [2] and [76] exploited two datasets, namely, UNSW-NB15 and CICIDS2018. In [2], the CICIDS2018 dataset produces impressive results with an accuracy of 99.620%. However, the UNSW-NB15 dataset leads to a much lower accuracy of 68.764%, which shows a significant gap in performance between the two datasets. In [76] the accuracy enhancement under poisoning assaults attains 48% on the UNSW-NB15 dataset and 36% on the CICIDS2018 dataset. The researchers produced a high value for performance metrics [14], [77].

The work in [78] used an autoencoder (AE) and evaluated CSE-CIC-IDS2018, USTC-TFC2016, and CIC-IDS2017. Compared with those obtained by GAN-based models, the F1-score is up to 89.27%, the accuracy is up to 91.13%, and the FDR is as low as 0.06. Other works, such as [58], achieved high threat detection and privacy preservation efficiency, with evaluations conducted on multiple datasets. Meanwhile, [99] reported an accuracy of 98.62, a precision of 98.90, a recall of 98.45, and an F1-score of 98.71. The study in [79] used CIC-IDS-2017 with an Accuracy of 97.1% and [80] Accuracy 98.62, Precision 98.90, Recall 98.45, F1-score 98.71, and in [81] the F1-score is approximately 3% for label-flipping attacks and 20%–64% for backdoor attacks.

IIoTset and InSDN were used in [82], and [58] used several datasets such as InSecLab-IDS-2021, Kitsune, CICIDS-2017, and InSDN. Both works focused on DDoS attacks in SDN-based networks.The dataset utilized in [83] is Modbus-based, which is important due to the widespread use of the Modbus protocol in industrial settings. The study employed several evaluation metrics and obtained an accuracy of 99.5%, along with high cross-validation results. The findings demonstrate that the FL-based method attains superior accuracy and reduced false alarm rates compared with traditional centralized ML models. The study in [84] used emulation and the CICEV2023 dataset. The dataset includes data from a health monitoring IoT system, which gathers vital health information using a Max30100 heart rate sensor. The research underlines the significance of data quality and balance in developing effective IDSs for IoMT systems.

Several key findings emerge based on the previous analysis of datasets used in existing FL-enabled IDSs. Notably, 34 studies rely on a few well-established benchmark datasets, including NSL-KDD, CIC-IDS2017, CICIDS2019, and CSE-CIC-IDS2018. By contrast, only a small percentage of research has used other datasets, such as the Modbus Network Dataset, MNIST, and CICEV2023 DDoS. This significant dependence on standard datasets imposes critical constraints on the generalization, robustness, and real-world applicability of FL-based IDS models. These datasets are mostly static, balanced, and gathered under controlled laboratory test conditions. Therefore, they fail to reflect actual IoT environments' non-IID, heterogeneous, and dynamic characteristics [105]. Research should prioritize using and developing datasets that more accurately capture the complexity of heterogeneity to promote generalizability in IoT networks and advance this field.

TABLE 4. Comparison of the specific IoT dataset used for FL-based IDS in IoT.

Table 4 Comparison of the specific IoT dataset used for FL-based IDS in IoT

Dataset	Year	Type of data	Availability	Number of Features	Samples Benign/Malicious		Attack type	#of Devices	FL-based IDS in IoT using dataset
(N-BaIoT) [40]	2018	Testbed	Yes	115	502595	6506674	BashLite Bot, Mirai Bot (Scan, TCP, UDP, COMBO, Syn, UDP Flooding).	9	[85],[86],[71],[87],[88]
BoT-IoT[42]	2019	simulated	Yes	46	9543	73360900	DoS, DDoS, service Scan, OS Fingerprinting, data theft, keylogging	5	[89], [90],[85], [14]
MQTTset dataset[43]	2020	simulated	Yes	34	11915716	165463	DoS, MITM (Flooding, Brute force, MQTT Publish flood, SlowIte, malformed data)	8	[91],[92], [93]
Edge-IIoTset[44]	2022	Real	Yes	61	11223940	9728708	UDP flood ,HTTP flood DDoS, TCP SYN DDoS, DDoS, Port ICMP OS Fingerprinting, flood DDoS ,scanning, Vulnerability scan attack, DNS spoofing, ARP spoofing, XSS, Backdoor, SQL Injection, Password cracking, Ransomware	10	[94],[72],[82],[95]
TON-IoT network t[45]	2021	simulated	yes	44	245000	124619	Backdoor, Data Injection, MITM, DoS, DDoS, Password cracking, Ransomware, Scanning, XSS	10	[90],[86],[73],[96],[97],[98],[99]
Kitsune [46]	2018	simulated	Yes	115	700000	26472754	DoS, Reconnaissance, botnet Malware, and Man in the Middle	9	[58]
Human Activity Recognition (HAR) [47]	2016	collected via smartphones	on request	565 frequency and time domain variables	2,365 recordings -	N/A	Internal and external threats, including activities such as walking, running, sitting,	N/A	[100]
NF-ToN-IoT-v2[48]	2020	simulated	Yes	44	6,099,469	10,841,027	Backdoor, DoS, DDoS, Injection, Password, MITM, Ransomware, XSS, Scanning	N/A	[101]
real urban IoT [49]	2021	Real	Yes	4,06	N/A	N/A	DDoS Attack	N/A	[102]
CIC-IoT22[50]	2022	Real	yes	84	243 746	232 365	RSTP, Brute Force, HTTP Flood, TCP Flood, and UDP Flood	60	[103]
WSN-DS [51]	2016	simulated	yes	23	374,661 Records		DDoS (Blackhole, grey hole attack, flooding, scheduling attack)	N/A	[104]

B. RQ3: WHAT ARE THE MOST RECENT TECHNIQUES UTILIZED TO CREATE AN IDS CAPABLE OF DETECTING ATTACKS IN IoT DEVICES?

The growing proliferation of IoT devices has brought significant benefits but has also heightened security vulnerabilities, particularly in DDoS attacks. These attacks flood systems with streams of unauthorized traffic, which leads to system downtime and financial losses.

IDSs are critical for securing IoT networks. However, conventional techniques often encounter obstacles due to the intricacy and scale of these environments. Recent advancements in ML and DL have been integrated into FL-enabled IDSs to overcome these limitations, significantly improving detection capabilities.

The advantages of ML and DL techniques include analyzing large-scale network data, adapting to threats, and identifying attack patterns (especially DDoS attacks).

This research explores the latest ML and DL techniques to develop FL-enabled IDS systems that effectively identify attacks on IoT devices. The literature on each approach is reviewed, and the effectiveness of ML and DL of FL-enabled IDS is evaluated in the subsequent sections. We categorize learning approaches into four types: supervised, unsupervised, semi-supervised, and reinforcement.

Figure 8 shows the classification of prevalent ML algorithms used in FL-enabled IDS.

1) SUPERVISED ML APPROACHES

a: DEEP LEARNING

A labeled dataset with preset input-output pairings is used to train a model [106]. This approach enables the algorithm to learn patterns from the data and develop a regression or classification model, which can subsequently be applied for real-time detection of DoS and DDoS attacks.

I) FEED FORWARD NEURAL NETWORK (FFNN)

The study in [103] introduced a novel DL-based IDS that is tailored to adapt to the growing security challenges of IoT networks. The proposed system leverages three DL techniques: feed-forward neural network (FFNN), random neural network (RandNN), and long short-term memory (LSTM). The system was rigorously evaluated using the CIC-IoT22 dataset. Among these techniques, the FFNN model demonstrates exceptional capability in identifying complex patterns; thus, it achieves superior performance to the LSTM and RandNN models. It also outperforms conventional ML approaches, such as K-nearest neighbor (KNN), naive Bayes, decision tree (DT), and random forest (RF). These findings underline the transformative impact of DL in IDSs, which suggests a promising route for future research and practical applications in securing complex networks.

Similarly, the work in [70] explored different neural network configurations to optimize DDoS detection using an FFNN with multiple hidden layers. The studies indicate that combining advanced neural networks with FL can significantly improve IoT security. They also present a platform for future research to further enhance these techniques for increasingly complex network challenges.

The research in [87] provided an FL-based IDS with locally adapted lightweight local learners to address IoT device restrictions. The authors combined supervised and unsupervised learning using a stacked AE and an FNN.

Experiments on the NBaIoT and CICIDS-2017 datasets indicate similar outcomes to centralized learning on metrics involving accuracy, recall, and precision while resolving privacy and data leakage issues. The author in [85] presented SparSified Federated Aggregation, which is a peer-to-peer FL method to reduce communication costs. It utilizes an FFNN with three hidden layers and offers a technique to fight against poisoning attempts over any random network topology. The method produces a false negative rate of less than 5% and an F1-score above 95%.

II) CONVOLUTIONAL NEURAL NETWORKS (CNN)

An FL-based IDS was proposed in [92] to boost the security of agricultural IoT infrastructures. The approach employs three types of classification techniques, namely, deep neural networks (DNNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs), to recognize various attacks, including DDoS efficiently. DNNs are effective for handling large datasets, CNNs are suited for analyzing spatial data patterns, and RNNs, especially LSTM networks, excel at processing sequential data. The framework's effectiveness was evaluated using CSE-CIC-IDS2018, MQTTset, and InSDN. These datasets cover multiple attack types. The findings reveal that the proposed system surpasses conventional centralized ML (non-FL) approaches in safeguarding the privacy of IoT devices while achieving the highest accuracy in detecting attacks. The study in [107] also leveraged CNN and FL in conjunction with Dempster–Shafer's theory to perform set-valued categorization of DDoS attack traffic in an industrial IoT (IIoT) setting. The pooling and convolution layers were combined in their evidence-based FL classifier (i.e., EFLDDoS) to extract high-dimensional features from local data belonging to private industrial clients. These studies illustrate the dynamic landscape of IDSs in IoT networks, emphasizing the critical role of advanced ML techniques in addressing cyber threats. Recent research has focused on improving IDSs for IoT networks by exploring advanced neural network architectures that enhance anomaly detection. The work in [14] described a significant approach combining LSTM networks with one-dimensional CNN (1D-CNN). This system effectively identifies anomalous data within benign traffic. It utilizes 1D-CNN to extract characteristics from sequential network data, which enhances classification accuracy in multi-categorical cyber-attack scenarios.

As cyber hazards develop in IIoT contexts, effective detection systems are important for safeguarding critical infrastructure. The research in [95] introduced a novel strategy

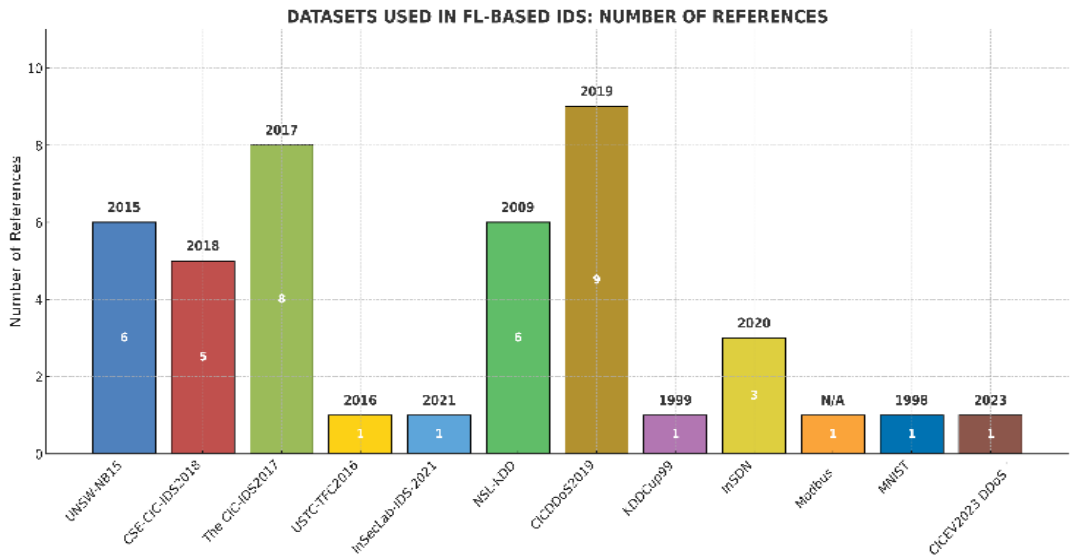

FIGURE 7. Published Studies are distributed according to the year of the dataset.

for identifying cyber-attacks using temporal convolutional networks (TCNs) within a federated sequential learning (FSL) architecture. This study evaluated various datasets, including MNIST, Bearing Defect Detection, Edge-IIoTset, and TON-IoT. The results showed that the FSL model consistently outperforms traditional models, such as LSTM networks and CNNs, regarding precision, recall, accuracy, and F1-score. The study emphasizes the potential of merging advanced neural network architectures with FL to improve intrusion detection.

The study in [2] explored advancements in anomaly detection systems by integrating several ML and DL approaches into FL frameworks to enhance IDSs. The authors combined DNN, GRU, CNN, and RNN to create robust IDS solutions. These techniques were complemented with FL strategies and privacy-preserving mechanisms to ensure effective and secure detection. The models were trained and assessed utilizing the UNSW-NB15 and CICIDS2018 datasets to offer a robust performance evaluation basis. The research highlights the potential for improving DDoS detection capabilities and sets the stage for future research to refine these techniques against evolving cyber threats in IoT environments.

Recent studies in cyber-attack detection are leveraging the combined strengths of various neural network models to improve detection ability, especially for DDoS attacks. The researchers in [67], employed a hybrid model that combines CNN for extracting features with the architecture of a multi-layer perceptron (MLP), which is known for its ability to learn complex nonlinear relationships through its interconnected layers of neurons. While MLPs excel at processing non-sequential data, they are not designed to process temporal relationships present in time-series information, which are better handled by RNNs.

As the need for robust IDS in IoT networks grows, several research studies have presented unique frameworks and strategies for improving security. The study in [76] demonstrated a comprehensive framework that integrates model- and data-level defenses, significantly improving detection accuracy. The SecFedsNIDS framework demonstrates significant performance improvements. It achieves up to a 48% increase in accuracy on the UNSW-NB15 dataset and a 36% improvement on the CICIDS2018 dataset, which are attributed solely to model-level defenses. In addition, data-level defenses contribute an extra 13% boost in accuracy on the CICIDS2018 dataset.

III) MULTI-LAYER PERCEPTRON (MLP)

The work in [86] aims to overcome inherent challenges, such as gradient vanishing, that can affect the learning capabilities of MLP. For this purpose, the MLP model was enhanced using residual connections, which help maintain learning effectiveness throughout deeper network structures. In addition, the model incorporates factorized convolutions to improve the efficiency of parameter aggregation. Another study [69] built on this foundation within an FL anomaly detection framework. This research presents details of an MLP framework that includes multiple dense layers. The framework was validated using the CIC-DDoS2019 dataset, which is explicitly designed to manage the imbalanced nature of DDoS attack data. It dynamically adjusts the training process based on the varying performance of individual clients, which tailors the model to predict better and mitigate potential attacks.

These experiments indicate the possibility of using integrated neural network models combining CNNs with MLPs to produce highly robust and adaptive IDSs. This approach fosters further research and development efforts to

TABLE 5. Comparison of the general IoT dataset used for FL based IDS.

Dataset	Year	Type of data	Availability	Data label	Number of Features	Samples Benign/Malicious		Attacks	FL-based IDS using the dataset
UNSW-NB15[53]	2015	simulated	**yes**	✓	49	2218761	321283	Worms, Fuzzers, Analysis, Shellcode, Backdoors, DoS, Exploits, Generic, and Reconnaissance	[77],[71],[2],[90],[76], [14]
CSE-CIC-IDS2018 [54]	2018	Testbed	yes	✓	83	7373198	1,019,203	DoS, DDoS, Heartbleed, Injection, DoS, Bruteforce (SSH, FTP), Scan.	[78],[92],[2], [90],[76]
The CIC-IDS2017 [54]	2017	Testbed	yes	✓	80	97718	128027	DoS, DDoS, Heartbleed, Botnet, Brute force, XSS, SQL Injection, Scan, Data exfiltration	[79],[80],[78], [58],[87],[74],[81],[104]
USTC-TFC2016 [56]	2016	Real	yes	✓	21	345407	406633	DDoS, DoS, Botnet, Brute Force, Web Attacks, Infiltration	[78]
InSecLab-IDS-2021[58]	2021	simulated	yes	✓	80	1449212	2642164	DDoS (UDP Flood, SYN Flood, ACK Flood)	[58]
NSL-KDD[59]	2009	simulated	yes	✓	42	67343	128682	DoS, Probe, R2L, and U2R.	[72],[71],[73],[74],[75], [14]
CICDDoS2019[60]	2019	Testbed	yes	✓	80	61000	45939000	DDoS attacks over the following protocols: NTP (DDoS NTP), NetBIOS (DDoS NetBIOS), DNS (DDoS DNS), LDAP (DDoS LDAP), MSSQL (DDoS MSSQL, SSDP (DDoS SSDP), SNMP (DDoS SNMP), UDP (DDoS UDP), UDP-Lag, Web DDoS, SYN and TFTP	[74],[108],[70],[107],[67],[109],[68],[69],[98]
KDDCup99 [52]	1999	simulated	N/A	✓	42	972781	3925650	N/A	[71]
InSDN [61]	2020	Real	yes	✓	76	68424	275,515	web attacks, DDoS, Botnet, DoS, password brute-forcing, probe, exploitation	[58],[92], [82]
Modbus network dataset[62]	N/A	N/A	yes	✓	34	N/A	N/A	DoS, Replay, MITM, Reconnaissance, Command Injection	[83]
MNIST [63]	1998	N/A	yes	N/A	784	N/A	N/A	N/A	[95]
CICEV2023 DDoS Attack Dataset[64]	2023	simulated	No	N/A	N/A	N/A	N/A	(DDoS) attack, DoS attack, Man-in-the-Middle (MitM)	[84]

refine these techniques, ultimately enhancing cybersecurity defenses in challenging modern network environments. This integrative method promises significant advancements in the field, which sets a precedent for future innovations in cybersecurity technologies. A study in [72] implemented an MLP model trained across multiple mobile edge computing servers to ensure distributed data handling. This model was validated using the NSL-KDD and Edge-IIoTset datasets. The model exhibits superior performance in accuracy and F1-scores to centralized models, such as support vector machines (SVMs) and RF. Thus, it enhances IoT attack detection.

IV) GATED RECURRENT UNITS (GRU)

GRUs are increasingly recommended for identifying anomalies in IoT networks due to their streamlined architecture and superior computational efficiency to LSTM networks. GRUs use fewer gates, which results in lower computational costs while often maintaining similar performance levels. In [83], GRUs and LSTMs are integrated within an FL framework, which utilizes ensemble learning techniques to enhance model performance. The study in [109]

compared a FedAvg-based method—which uses two GRU layers and two fully linked layers activated by ReLU—to one that utilizes LSTMs and typical weights. The results revealed that the federated IDS (FIDS) beats previous methods in terms of convergence rate and stability, which demonstrates the effectiveness of these neural network architectures in enhancing anomaly detection. Overall, these studies illustrate the potential of GRUs and ensemble learning inside FL frameworks to confront the issues of cyber threats in IoT contexts and point to areas for further studies and practical implementations.

The researcher in [97] proposed an FL-based anomaly detection system for identifying IoT network assaults. This method employs numerous GRU layers to improve accuracy, precision, recall, and F1 scores. The ensemble approach, which combines predictions from multiple GRU layers, enhances performance. Experiments demonstrate that this strategy outperforms previous techniques for securing user data while optimizing statistical analysis, energy economy, and memory utilization. LSTM and GRU-NN models were trained on Modbus data, which results in effective intrusion detection in IoT.

The author in [68] developed a hybrid RNN model. The model combines LSTM, BiGRU, and BiLSTM to analyze the CICDDoS2019 dataset. This technique achieves outstanding results, with an accuracy of 99.59%, an FPR of only 0.042%, and a detection time of 0.062 ms.

V) LONG SHORT-TERM MEMORY (LSTM)

The work in [108] introduced lightweight residual networks, which balance computational efficiency with high accuracy. These capabilities make them particularly appropriate for resource-constrained IoT devices. This research employed DL models, such as deep CNNs, to identify several types of harmful traffic and LSTM networks to classify DDoS attacks. These techniques demonstrate effectiveness but present trade-offs in computational resource requirements and training times. The study in [102] further explored the capabilities of LSTM by proposing a detection framework utilizing fog and cloud layers. The framework was evaluated on a real urban IoT dataset to detect DDoS attacks.

VI) DEEP BELIEF NETWORK (DBN)

The work in [99], proposed a Pelican Optimization Algorithm with FL Driven Attack Detection and Classification (POAFL-DDC) approach. This method follows the deep belief network (DBN) model for attack detection. It then employs POA to optimize the DBN hyperparameters. Furthermore, the TON_IoT dataset was employed to measure experiment validity for the POAFL-DDC algorithm and tested against several metrics. The results showed that the POAFL-DDC strategy was superior to that of the remaining models.

VII) RECURRENT NEURAL NETWORKS (RNN)

The research in [2] presented advancements in anomaly detection systems by integrating several ML and DL approaches into FL frameworks. The authors combined DNN, GRU, CNN, and RNN to develop IDS solutions. These techniques were complemented with FL strategies to ensure security and privacy detection.

Recent advancements in IDSs have underscored the need for effective solutions in IoT networks, where privacy and security are paramount. A study in [94] proposed an FL method that is specifically designed to detect intrusions while ensuring the confidentiality and security of local device data. In this approach, clients locally train their models on-device and only exchange parameter updates with a main server, which aggregates them to improve the detection algorithm. The model obtains an accuracy of 92.49% when tested on the Edge-IIoTset dataset, which closely approaches the accuracy of conventional centralized ML models at 93.92%.

VIII) ARTIFICIAL NEURAL NETWORK (ANN)

Recent advancements in cyber-attack detection showcase the promise of artificial intelligence in enhancing security. The research in [98] utilized an artificial neural network model trained on labeled data to identify cyber-attacks. This model achieves great accuracy and few false positives by merging features from multiple datasets, including ToN_IoT and CICDDoS2019. Meanwhile, the study in [77] introduced the FogFed framework, which employs a two-tier detection system: a fog-based IoT attack detection mechanism using a binary FL classification and a cloud-based multiclass classifier. This framework utilizes DL architectures via libraries, such as PyTorch, although specific models are not mentioned. The authors compared FogFed with several advanced ML and DL models, including ensemble SVM, CNN, and ensemble statistical analysis. The comparison highlights the innovative nature of FogFed and the ongoing need for effective IoT security solutions. Collectively, these studies significantly

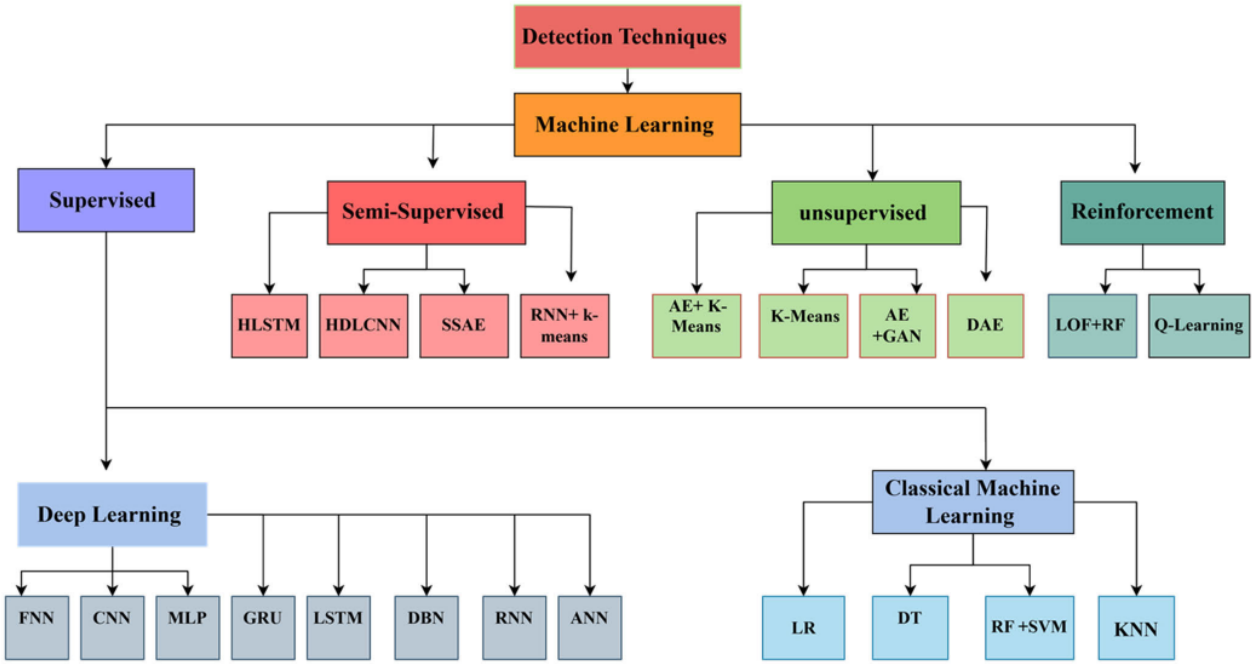

FIGURE 8. Detection approaches taxonomy.

advance the field of cyber-attack detection by addressing potential and obstacles to implementing advanced ML techniques in real-world applications.

The study [75] used a neural network to classify attack samples in IoT intrusion detection, with a training set developed through the negative selection algorithm to enhance classification robustness. The FL strategy beats device-trained models and reaches accuracy levels comparable to centralized models when evaluated using the NSL-KDD dataset. This performance demonstrates the potential of FL to maintain high detection accuracy while protecting data privacy, which is vital to IoT applications. The scheme proposed by the authors improves detection accuracy and offers efficiency and scalability, which marks a significant contribution to addressing key challenges in intrusion detection while prioritizing user privacy.

The author in [93] presented an FL-based IDS approach with multi-view information of IoT network data to identify and classify intrusions in a decentralized format. Multi-view ensemble learning enables the system to achieve enhanced learning efficiency across diverse attack vectors. The FL component performs the prole aggregation effectively with the added benefit of peer learning. Meanwhile, it detects the assault with high accuracy, as opposed to the conventional ML algorithms using a single feature set of attack information in a decentralized manner.

b: CLASSICAL MACHINE LEARNING

I) LOGISTIC REGRESSION (LR)

The work in [91] assessed various classifiers using healthcare datasets, including RF and KNN. Notably, RF is proven to be the most accurate, which reaches a peak of 99.69%. This finding suggests its robustness in handling complex data structures typical in healthcare settings. The study explores advanced ML techniques in cybersecurity, albeit with different approaches and outcomes.

II) DECISION TREES (DT)

The research in [82] presented an FL model called F-BIDS, which integrates DTs and RF classifiers to generate metadata subsequently used to train a neural network meta-classifier. This model performs exceptionally on the Edge-IIoT and InSDN datasets, particularly in multi-class classification tasks. It outshines standard models such as SVM and MLP. This result indicates the ability of FL frameworks to improve privacy-preserving data analysis across distributed networks. Overall, these studies underscore the effectiveness of customized ML solutions in cybersecurity.

The most recognized instance of this network type is illustrated in [89], an innovative intrusion IDS employing fog computing to identify DDoS attacks in blockchain-enabled IoT networks. The IDS utilizes RF and DTs using XGBoost, which is a scalable ML algorithm. Gradient boosting algorithms are extremely efficient in speed and performance, which makes them ideal for real-time DDoS detection.

In addition, DT is proposed in [110] to implement an FIDS for IoT environments. This approach leverages the distributed nature of IoT devices to create a scalable and privacy-preserving IDS.

III) RANDOM FOREST (RF)

RF is a popular ensemble learning method in ML, and it is commonly applied to classification and regression tasks.

Compared with a single DT, RF has a lower risk of overfitting and is more robust to outliers. However, RF can be computationally expensive and memory-intensive, particularly when working with huge datasets and trees. The authors in [84] utilized an enhanced RF model for attack classification, which achieves an exceptional accuracy of 99.98%. In the subsequent stage of anomaly detection, an SVM was employed, demonstrating a high accuracy of 99.7% in identifying anomalies. The suggested system was evaluated using the CICEV2023 dataset, which effectively detects and categorizes various types of attacks.

IV) K-NEAREST NEIGHBOR (KNN)

KNN is a supervised ML algorithm. It uses distance measures to classify data points or predict continuous values in regression tasks [111]. The author in [14] combined different techniques using KNN with 1D-CNN, LSTM, and neural networks to detect attacks. The method achieves an accuracy of 99.12%, reduces average loss, and improves execution time. Table 6 includes a concise description of the advantages and disadvantages of the techniques with the most relevant approaches, focusing on supervised techniques in existing FL-enabled IDS approaches.

2) SEMI-SUPERVISED LEARNING

Semi-supervised learning integrates labeled and unlabeled data and improves the performance and generalization of ML models, especially in cybersecurity for IoMT scenarios. Table 7 concisely describes the advantages and disadvantages of semi-supervised techniques in existing FL-enabled IDS approaches.

a: HIERARCHICAL LSTM (HLSTM)

A few works have employed variations of such techniques to develop FL-based IDSs. Examples include TL utilizing AEs. The research in [73] suggested an IDS based on the hierarchical LSTM model to process sequential data generated by IoMT efficiently. The system, which was implemented on distributed Dew servers utilizing cloud computing, achieves an accuracy of 99.31% and minimal training loss with the TON-IoT and NSL-KDD datasets. This method demonstrates the efficacy of combining hierarchical models with distributed computing for real-time cyber-attack detection. However, additional research is required to evaluate scalability and adaptability in various environments.

b: HIERARCHICAL DEEP LEARNING CONVOLUTIONAL NEURAL NETWORK (HDLCNN)

The authors in [100] explored the approach by combining an artificial neuro-fuzzy interface system (ANFIS) with a hierarchical DL CNN (HDLCNN) to enhance intrusion detection capabilities. The HDLCNN component is primarily employed for initial data classification, effectively segregating relevant data. Subsequently, ANFIS uses fuzzy logic and neural networks to adapt and refine detection rules. The hybrid approach also significantly reduces the FPR, which addresses key challenges in IDS performance metrics.

This integration showcases the potential for neural networks and fuzzy systems to work collaboratively to create highly reliable and efficient IDSs.

c: STACKED SPARSE AUTOENCODER

In [80], the authors presented a comprehensive approach combining stacked sparse AE for feature reduction, TCN for temporal pattern recognition, and FL for collaborative and privacy-preserving model training. This combination solves the difficulties of limited resource availability on edge devices, data privacy, and the necessity for rapid and reliable intrusion detection in IoT environments. The effectiveness of this system was assessed using the CIC-IDS-2017 dataset, in which it produces high-performance metrics: an accuracy of 98.62%, precision of 98.90%, recall of 98.45%, and an F1-score of 98.71%. This result highlights the power of integrating these methods to achieve rapid and accurate intrusion identification while protecting data privacy.

d: RECURRENT NEURAL NETWORKS AND K-MEANS

The authors in [74] integrated advanced ML and DL techniques within an FL framework to offer a secure and effective IDS for recognizing DDoS assaults in IoT contexts. The authors employed an AE for feature extraction and dimensionality reduction by utilizing FL, RNN, K-means clustering, and SMOTEENN. This approach aids in diminishing the computational cost and enhances the RNN model's performance. The authors used K-means clustering to categorize clients with similar data distributions, which facilitates hierarchical model aggregation and diminishes the number of communication cycles necessary for model convergence within the FL framework. Moreover, with the implementation of edited nearest neighbors (SMOTEENN), the latent space is regularized through adversarial learning. VAEs map input data to a distribution to improve the capacity of the model to capture underlying patterns. These developments improve the performance of AEs in IDSs. The techniques were evaluated on multiple datasets, including CSE-CIC-IDS2018, USTC-TFC2016, and CIC-IDS2017, which show high detection accuracy and low false discovery rates.

3) UNSUPERVISED ML APPROACHES

Unsupervised ML techniques offer a promising approach to enhancing IDS by identifying anomalies without needing labeled datasets. These methods, including K-Means clustering, PCA, and Autoencoders are particularly effective in detecting DDoS attacks within dynamic IoT networks. By analyzing unusual behavior patterns, unsupervised ML can identify novel threats not present in the training data. By analyzing unusual behavior patterns, unsupervised ML can identify novel threats not found in the training data. This adaptability is critical for managing the changing threat landscape of IoT systems. Overall, unsupervised ML significantly strengthens IDS capabilities in detecting sophisticated cyber

TABLE 6. Summary of the supervised approach used for FL to enable IDS.

ML detection approach	Approach	Technique	Description	Advantage of Technique	Disadvantage	Works using this technique
Supervised	Deep Learning	Feedforward Neural Network (FNN)	Feedforward neural network	Simple architecture, effective for many tasks	Prone to overfitting, not well-suited for sequential data	[85], [70, 103],[87]
		Convolutional Neural Networks (CNN)	The deep learning model for spatial data	It captures spatial features effectively and is suitable for image and grid-like data.	Requires large datasets and significant computational resources.	[94], [2], [92], [14], [95], [76], [107], [67]
		Multilayer Perceptron (MLP)	Feedforward artificial neural network	Simple and effective for a wide range of problems, good for non-linear mappings	Limited in handling sequential data, can overfit if the network is too complex	[86], [72], [67], [69]
		Gated Recurrent Units (GRU)	Gated Recurrent Units	Efficient and less complex than LSTMs for sequence modelling	Still computationally expensive, can be less interpretable than simpler models	[58], [2], [68], [83], [97],[68]
		Long Short-Term Memory (LSTM)	Long short-term memory network	Effectively addresses the vanishing gradient problem, making it proficient in handling long-term dependencies in sequential data.	Computationally expensive, requires significant memory and processing power	[58], [14], [102], [103], [108], [109], [68], [83] [112]
		Deep Belief Networks (DBN)	Deep Belief Networks	Effective for deep architectures	Complex to train, computationally intensive, and requires large datasets	[99]
		Recurrent Neural Networks (RNN)	Sequential data model	Effective for sequential data; captures temporal dependencies	Prone to vanishing gradient issues, complex to train.	[94], [103], [2], [92]
		Neural Networks (ANN)	General neural network	Versatile and powerful for modeling complex data	Requires extensive tuning and large datasets, difficult to interpret	[14], [82], [75], [98],[77],[93]
	Classical Machine Learning	Logistic Regression (LogR)	Statistical model for binary classification	Simple and interpretable, effective for binary classification problems	Assumes linear relationship between features and target, can perform poorly with complex relationships	[91]
		K-Nearest Neighbors (KNN)	Instance-based learning algorithm	Simple and effective, easy to implement and interpret	Computationally intensive during prediction, performance decreases with high-dimensional data	[14],[91]
		Adaboost (AB)	Ensemble method	Combining weak classifiers to create a strong classifier reduces bias and variance.	Sensitive to noisy data, can overfit if the base classifiers are too complex	[91]
		Decision Tree (DT)	Decision Tree	taking care of both numbers and categorical data	Susceptible to overfitting, capable of generating complex trees that lack effective generalization.	[89], [91], [82], [110]
		Support Vector Machines (SVM)	Support Vector Machines	Effective for high-dimensional spaces, robust to overfitting	Not suitable for very large datasets, it can be sensitive to the choice of kernel and regularization parameters	[84]

threats. Table 8 briefly explains the advantages and drawbacks of the unsupervised techniques in FL-enabled IDS.

a: AUTOENCODERS AND K-MEANS CLUSTERING

The authors [71] discuss a novel collaborative learning framework for FL-based IDS to identify DDoS attacks in IoT networks. It integrates FTL and Unsupervised Deep Learning with autoencoders and k-means clustering to handle heterogeneous and unlabeled data effectively. This method improves anomaly detection and network behavior classification, considerably increasing the detection accuracy and efficiency of IDS. This innovative framework highlights advancements in applying complex ML strategies to strengthen IoT security.

b: K-MEANS CLUSTERING

K-means clustering plays a critical role in identifying malicious participants in IDS. Study [79] describes a method that uses K-means to detect outliers, calculating Manhattan similarity of loss scores to identify potential malicious activity. This approach preserves the integrity of the global model by filtering out harmful local models. The study [113] used FL to train models while preserving data privacy cooperatively. Device heterogeneity was addressed using a clustered FL strategy that included K-means and PCA, while anomaly detection was done using unsupervised autoencoder models trained mostly on benign traffic data.

c: AUTOENCODERS AND GENERATIVE ADVERSARIAL NETWORKS

IDS utilize autoencoders because of their ability to encode and decode data, learning informative representations. Simple autoencoders use completely connected layers to reconstruct data with as little loss as possible. Advanced versions such as Variational Autoencoders and Adversarial Autoencoders, are examined in the study [78]. AAEs use Generative Adversarial Networks to regularize the latent space through adversarial learning. VAEs map input data to a distribution to improve the model's capacity to capture underlying patterns. These developments improve autoencoders' performance in IDS. The techniques were evaluated on multiple datasets, including CSE-CIC-IDS2018, USTC-TFC2016, and CIC-IDS2017, and showed high detection accuracy and low false discovery rates.

d: DEEP AUTOENCODER (DAE)

The work in [90] proposed an unsupervised DAE model for anomaly detection tasks. The authors described a stacked-unsupervised federated FL approach that combines DAEs with an energy flow classifier to improve intrusion detection across various network systems. This cross-silo FL model leverages shared knowledge from multiple participants to create a unified and generalized IDS solution. This way effectively improves its adaptability to various network environments. Related to this work, the study in [88] proposed an FL combined with DAEs and traditional ML techniques to create a robust and privacy-preserving IDS for IoT environments. This system can effectively identify DDoS attacks. In a recent work, researchers employed a multifaceted approach within an FL framework to improve the effectiveness of IDSs against attacks in IoT environments.

4) REINFORCEMENT-LEARNING (RL)

RL is an ML approach technique where an agent interacts with an environment to acquire decision-making skills. Then, it continuously adjusts its actions to optimize cumulative rewards. This learning process distinguishes RL from supervised learning, where models are trained on labeled datasets with predefined input-output pairs. By contrast, RL agents lack prior knowledge of the outcomes in the environment. Instead, they acquire knowledge via trial and error, which involves modifying their activities according to the feedback derived from the consequences of their decisions. This adaptive learning technique allows the agent to enhance its decision-making capabilities progressively, even in complex or dynamic environments. Table 9 shows a concise overview of the advantages and drawbacks of reinforcement techniques in present FL-enabled IDS technologies.

a: LOCAL OUTLIER FACTOR (LOF) AND REINFORCEMENT LEARNING (RL)

The study in [81] demonstrated that FL, when combined with advanced ML techniques such as local outlier factor and RL, can significantly enhance the capability of IDSs to identify DDoS attacks within IoT environments. The proposed two-phase defense mechanism, which is called DPA-FL, successfully mitigates the risk of poisoning attacks, with a notable accuracy rate of 96.5% in defense. Moreover, this approach significantly enhances the F1-score by 20%–64% under backdoor attacks compared with other defense mechanisms. This finding underscores the potential of integrating FL with sophisticated ML techniques to bolster security in IoT networks.

b: Q-LEARNING

The research in [96] employed the model-free RL algorithm to help systems learn optimum actions across different states for maximizing cumulative rewards. This technique is instrumental in adapting to cyber-attack's evolving and dynamic patterns. Then, the fusion aggregation strategy was implemented, which is pivotal in consolidating the knowledge from multiple local models into a robust global model. This method guarantees that the IDS leverages all varied facts and experiences of contributing nodes, which leads to a more precise and comprehensive detection system. The study used modern IoMT datasets, in which the proposed method obtains an impressive accuracy rate of 99.40%, a detection rate of 98.99%, and a reduction in communication overhead by 57%. These results highlight the efficacy of combining RL with aggregation strategies to enhance the performance of IDSs in modern cyber-attack scenarios.

5) DISCUSSION

The previous description and our analysis reveal that most works utilize supervised learning approaches. These methods typically perform well in detecting various cyber-attacks but require labeled data, which introduces major challenges in real-world applications. Considering the dynamic nature, scale, and diversity of current deployments, this requirement can be impractical in many scenarios, such as IoT, where numerous devices need this information to identify potential threats. Furthermore, supervised learning techniques cannot detect previously unknown attacks, another inherent limitation in most FL-enabled IDS approaches.

Regarding the considered techniques in supervised learning (Figure 9), LSTM is the most widely used approach (11 out of 43) either uniquely or in combination with other techniques, such as CNN+GRU [58], CNN+KNN+NN [14] or GRU [83]. Several researchers have also used the CNN technique and has also been utilized independently in some works. For example, in [76], the proposed approach uses CNN with two datasets from CICIDS2018 and UNSW-NB15. Using these datasets, the accuracy improvements attained are up to 48% on the UNSW NB-15 dataset and up to 36% for the CICIDS2018 dataset. Data-level defense may lead to an accuracy improvement of up to 13.4% on the CICIDS2018 dataset. Meanwhile, in [2], CNN, DNN, and GRU were used with the same datasets as in [88]. The obtained accuracies are 99.620% and 68.764% for the CICIDS2018 and UNSW-NB15 datasets, respectively. Notably, most of the analyzed works focused on specific evaluation metrics.

Most works primarily emphasize recall, precision, accuracy, and F1-score. However, the work in [67] used the CNN technique but differed from others by focusing on computational time (3.917 ms) and ROC AUC (0.9890). Moreover, although unsupervised and semi-supervised methods address some limitations of supervised learning, our survey revealed that only a small fraction of the studies utilize these techniques. Specifically, only 6 out of 43 FL-enabled IDS implementations use semi-supervised techniques, and only 4 out of 43 (or 9.3%) employ unsupervised approaches.

In the case of unsupervised techniques, AE is the most widely known. However, the surveyed works use simple or stacked AEs. For example, only one exploits VAE (AAE), GAN, and BiGAN [78], which are considered promising intrusion detection approaches. Furthermore, GANs have been used to generate attack data that could help improve the effectiveness of developed approaches. Owing to the recent advances in generative models, these techniques will likely expand in the future years for the advancement of IDS.

Finally, only 4.65% (2 out of 43) of the analyzed studies utilize an RL approach. Despite its potential advantages in adapting to environmental changes for attack detection, RL remains underutilized. Recent research [96] highlights the benefits of using RL in IoMT networks by demonstrating its adaptability, accuracy, and efficiency in detecting cyber-attacks. However, challenges such as data requirements, computational complexity, slow convergence time, and difficult deployment still need to be addressed. Therefore, we recommend using RL in FL to adapt to evolving threats. Combining various approaches and techniques often enhances the effectiveness of intrusion detection in many real-world scenarios. Therefore, we advocate combining DL strategies (e.g., CNN, LSTM, and RNN) with traditional models (e.g., RF, SVM, and KNN) to improve feature representation and decision-making among heterogeneous data. Furthermore, integrating supervised and unsupervised learning approaches combines the advantages of accurate classification and creative anomaly detection. This hybrid strategy enhances the generalizability and robustness of systems in diverse and privacy-sensitive contexts.

C. RQ4: WHAT ARE THE PRIMARY CHALLENGES AND LIMITATIONS FOR IMPLEMENTING FL-BASED IDS IN IoT APPROACHES?

FL presents a promising structure for designing IDSs in IoT environments. However, despite its promise, FL-based IDS implementations encounter significant challenges that need to be addressed to unlock their full potential. This section highlights the significant difficulties and outlines potential research directions.

1) DATA HETEROGENEITY

IoT networks comprise various devices with varying computational power, network capabilities, size, distribution, and quality [114]. This heterogeneity impacts training time and model accuracy and complicates the standardization of the FL training process because different devices contribute unequally to the global model. Devices with limited computational resources may struggle to handle ML tasks, while energy-constrained devices may experience dropouts, which ultimately compromises the consistency and quality of the IDS model. Data heterogeneity, also known as statistical heterogeneity, arises due to uneven data distribution among clients. This occurrence leads to a scenario where the data do not follow the same sampling pattern, which results in non-IID data. This phenomenon poses unique challenges in FL. Current studies address the challenge of data heterogeneity using various approaches. A device selection method is introduced to address system heterogeneity, where only devices with ample available resources are chosen to participate rather than involving all devices [115], [116]. Efficient client selection plays a pivotal role in enhancing communication efficiency and minimizing data heterogeneity [117]. Furthermore, clustering is a prominent approach for addressing data heterogeneity in FL. This technique involves grouping clients based on various similarities in attributes such as data distributions, local patterns, or model updates of different clients [118]. The study [119] presented a hierarchical clustering approach to address the challenges of

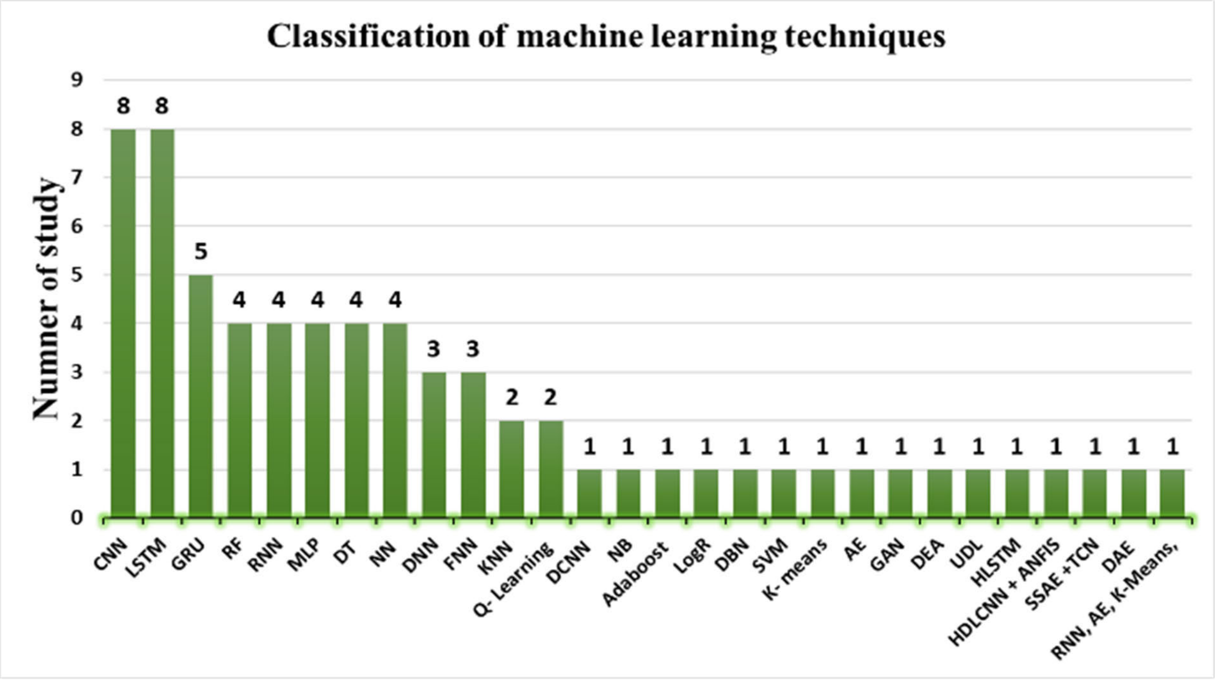

FIGURE 9. ML techniques by existing FL-enabled IDS.

statistical heterogeneity in FL, which restricts convergence and increases communication cycles.

2) PRIVACY AND SECURITY

Privacy and security risks remain key concerns in FL systems. FL aims to preserve privacy by keeping data localized on devices, but it remains vulnerable to attacks. Inference attacks allow adversaries to extract sensitive information from shared model updates, although raw data are not directly shared [120]. Multi-party computation (MPC) and differential privacy (DP) were used by the authors to address privacy threats [121], [122], [123]. These strategies include incorporating noise into sensitive attributes and applying Gaussian noise to local model gradients to protect personal information and facilitate model convergence. Specifically, the authors in [121] employed a method that combines homomorphic encryption (HE) with DP. Limited client participation in DP frameworks poses a critical challenge because low interaction may introduce distortions in aggregated models. This situation underscores a fundamental trade-off between privacy preservation and model accuracy in FL systems. To address this issue, novel normalization methods incorporating neural networks were used in [124]. Similarly, model poisoning attacks pose a substantial risk, where malicious participants intentionally corrupt the global model by submitting manipulated updates [125]. This condition not only deteriorates IDS performance but also compromises the overall security of the IoT network, especially in sensitive environments such as healthcare or industrial systems.

3) SCALABILITY IN HETEROGENEOUS IoT

Scalability poses a significant challenge for FL in IoT environments, given that the vast number of interconnected devices causes communication overhead [126]. Client selection was proposed in [127], [115], and [116], which helps to mitigate the communication overhead and latency issues arising from the diverse capabilities of IoT devices. Frequent model updates can overwhelm network bandwidth, which causes delays and reduces IDS performance [128]. Moreover, devices with restricted connectivity or insufficient bandwidth may encounter participation difficulties, leading to inconsistent model contributions [18].

4) DATA IMBALANCE

Data imbalance further complicates FL because IoT devices produce varying amounts of data. This complexity leads to skewed data distributions, which cause a model bias toward devices or classes with more data. Ultimately, the accuracy is affected by underrepresented devices. Addressing this imbalance is essential for ensuring comprehensive IDS performance across the entire network. Advanced techniques have been developed to address non-uniform data distributions and diverse data features [129], which reduces data bias in FL for IDS. Data enhancement methods generate synthetic data to correct distribution imbalances; meanwhile, TL leverages models trained on single nodes as a basis for others, effectively addressing data discrepancies [130]. Furthermore, federated meta-learning and adaptable FL adapt dynamically to heterogeneous data distributions, which enhances the learning process [131].

TABLE 7. Summary of the Semi-supervised approach.

ML detection approach	Technique	Description	Advantages of the Technique	Disadvantage	Works using this technique
Semi-Supervised	HLSTM	Hierarchical Long Short-Term Memory	Extends LSTM with hierarchical layers to capture more complex dependencies.	It is computationally intensive and requires significant resources.	[73]
	HDLCNN + Adaptive Neuro-Fuzzy Inference System (ANFIS)	Combines convolutional networks with neuro-fuzzy systems for better learning.	High accuracy in complex tasks, adaptive learning capability.	Complex to implement and train.	[100]
	Stacked Sparse Autoencoder SSAE +temporal Convolutional Network (TCN)	It uses autoencoders for feature extraction and TCN for temporal data processing.	Effective for capturing temporal dependencies and reducing dimensionality.	Computationally expensive.	[80]
	RNN, AE, K-Means	Combines Recurrent Neural Networks, Autoencoders, and K-means for robust learning.	Integrates temporal analysis, feature learning, and clustering.	High computational cost, complex integration.	[74]

TABLE 8. Summary of unsupervised techniques used for FL to enable IDS.

ML detection approach	Technique	Description	Advantages of the Technique	Disadvantage	Works using this technique
Unsupervised	DL (UDL) autoencoder and k-means techniques	It combines an autoencoder for feature learning with K-means for clustering.	Effective for feature extraction and clustering in unlabeled data.	It can be complex to train and requires careful tuning	[71]
	K- means	Clustering algorithm	Simple and scalable, effective for clustering large datasets	Assumes spherical clusters of equal size, sensitive to initial centroids, and can converge to local optima	[79],[113]
	Autoencoder (AE)	Autoencoder	Reduces dimensionality, effective for anomaly detection and feature extraction	It can be challenging to train, prone to overfitting on small datasets	[78]
	Generative Adversarial Network (GAN)	Generative Adversarial Network	Generates synthetic data, effective for data augmentation and anomaly detection	Difficult to train and can be unstable, requires careful tuning of hyperparameters.	[78]
	Deep Autoencoder (DEA)	Deep Autoencoder	can effectively reduce the dimensionality of data also Automatically identify and extract the most significant features from raw data, and is highly effective in anomaly detection tasks by identifying deviations	prone to overfitting, computationally intensive	[90], [88]

5) FAIRNESS

Fairness in FL ensures that the learning process and outcomes are distributed equally among all participating devices or nodes. Significantly imbalanced data distributions may create biases in the aggregated global model. Furthermore, inequalities in computational capabilities among devices may result in dominant contributions from more capable clients, except those with fewer resources. Addressing these challenges is necessary to ensure that FL delivers equitable benefits across IoT networks. Recent studies have focused on various strategies to enhance fairness in FL. For example, the study in [132] presented a method that enhances the target distribution of the model by using a mixture of client distributions. Moreover, the research in [133] explored fairness-aware client selection mechanisms that reduce communication overhead and maintain adaptive fairness in environments under resource limitations. Developing lightweight fairness solutions that can function efficiently in resource-limited IoT devices is crucial for ensuring equitable model performance.

6) RESOURCE CONSTRAINTS

IoT devices often operate under stringent constraints of computational, memory, and energy, which render the implementation of FL challenging. FL is computationally intensive and requires iterative local model training with frequent communication in a central server, which can strain these resource-constrained devices. Training complex ML models on low-power IoT nodes may result in significant energy consumption, reducing device longevity and compromising network sustainability [134]. Many IoT devices may lack the necessary computational power to engage in local training, which limits their contribution to the FL process [135]. Addressing these problems requires developing lightweight and efficient FL techniques that minimize computational demands, energy consumption, and communication overhead while preserving model accuracy [136]. Resource-constrained IoT devices can participate in FL networks through optimized strategies for reducing memory usage and data transmission [137].

7) COMMUNICATION OVERHEAD

The primary issue of any FL application is the cost of communication per round of training [138]. Federated training demands sending and receiving the model parameters (e.g., weights, batch size, and learning rate) from the server to the clients, and vice versa. This process can be computationally expensive and time-consuming. Statistical heterogeneity is a major contributor to communication complexity. Non-IID data distribution [18] might cause communication bottlenecks and additional costs. FL collects data locally on each device according to usage patterns and surroundings, underscoring the importance of understanding data distribution. Client selection has been proposed to reduce the communication overhead and latency caused by the diverse capabilities of IoT devices [115], [139]. Clustering clients reduces

TABLE 9. Summary of the Reinforcement approach used for FL- enable IDS.

ML detection approach	Technique	Description	Advantages of the Technique	Disadvantage	Works using this technique
Reinforcement	(LOF)+(RL)	Applies Local Outlier Factor (LOF) and Reinforcement Learning	LOF enables rapid anomaly detection, whereas RL dynamically alters thresholds for better accuracy and flexibility in challenging conditions	LOF suffers from borderline and is sensitive to parameters, but RL is computationally intensive and complicated to execute effectively.	[81]
	Q- Learning	Applies Reinforcement learning principles in a federated learning setting.	Enhances learning efficiency and scalability across distributed systems.	Requires significant computational resources and coordination.	[96]

communication costs by reducing the number of model parameters. Each group of clients communicates with the corresponding server [140]. Quantization solutions are also essential for minimizing transmitted data's size, and reducing communication delay overhead. This reduction is crucial for improving responses to the constraints of IoT devices for energy, computation, and memory [141], [142].

VI. FUTURE DIRECTIONS

FL could revolutionize IDSs by allowing the development of highly effective and efficient models that maintain data privacy. Although FL is still in its early phases of development, it holds significant potential for advancement in various promising areas. The following outlines some possible future directions for FL in IDSs.

A. MANAGING DATA HETEROGENEITY

Data heterogeneity can cause model divergence, slow convergence rates, and reduced accuracy in FL. Models trained on non-IID data may not generalize well across all devices, which leads to suboptimal intrusion detection performance [143]. Developing models tailored to individual devices or clusters of devices can enhance performance by accommodating specific data distributions. Clustered FL enables the creation of specialized models for each group, which improves overall detection accuracy [113]. Research should shift toward developing algorithms that can dynamically cluster devices based on real-time data characteristics to realize more responsive and effective management of data heterogeneity.

B. ENHANCING PRIVACY AND SECURITY

The implementation of FL-based IDSs in IoT environments demands robust privacy and security measures. Key privacy-preserving techniques are discussed as follows. Secure MPC (SMPC) is a prominent technique that allows collaborative processing without exposing private input data [144]. However, the approach faces a critical limitation in computational efficiency in execution time, especially when processing large-scale datasets or operating under real-time constraints. Another approach to guarantee data privacy is HE, which allows computations on encrypted data [145]. This field has been developed by integrating HE with differential privacy into IoT intrusion detection systems, effectively reducing the exposure of sensitive network patterns during federated analysis [146]. Security measures involve robust aggregation methods to mitigate malicious inputs [147], Anomaly detection to identify attacks [83], and secure communication protocols such as transport layer security to protect data transmission [148]. These measures collectively provide a layered defense system. Nonetheless, challenges such as computational overhead and compatibility between privacy and security methods need additional research into developing lightweight, adaptive hybrid systems [149]. Despite many works having been presented to address this issue, it still remains a complex challenge. Therefore, as part of future directions, the focus should be on integrating HE and SMPC to protect model updates without compromising training efficiency. In addition, future work must focus on developing robust aggregation algorithms resilient to data poisoning attacks. Federated anomaly detection should be utilized to identify malicious participants without central access to raw data. Another approach for optimizing privacy and developing comprehensive incentive structures is to integrate blockchain with FL. The hybrid integration allows data to remain on local devices and ensures that only encrypted model updates are shared [150], [151]. Furthermore, the detection of novel cyber threats risks throughout networks needs to be enhanced [152]. Future implementations should focus on exploring dynamic resource allocation mechanisms to effectively balance the computational and storage requirements of blockchain integration within the constraints of resource-limited FL.

C. IMPROVING SCALABILITY

Strategies must reduce communication overhead and minimize network bandwidth consumption as IoT devices grow to enhance scalability while ensuring efficient and reliable model updates. A crucial factor in this process is limiting the frequency of device-to-server communication because excessive interactions can significantly increase network traffic and latency. Future studies should aim to enhance methods that minimize data transmission during the learning process, such as compression strategies, quantization, and gradient sparsification, to reduce the size of model updates, which are vital for large-scale IoT networks [153]. Enhancing computational optimization should focus on using edge devices for localized data preprocessing and computing and reducing the load on central servers, which results in real-time scalability [154]. In addition, future research should focus on system architecture design using multi-layered topologies. In these topologies, edge devices coordinate local updates before global aggregation to reduce communication and computational load [44], [73].

D. ADDRESSING DATA IMBALANCE

Addressing data imbalance is crucial for establishing a successful FL-based IDS in IoT contexts [155]. Data imbalance arises when the distribution of classes in local or global datasets is uneven, which may decrease IDS performance and generalization [156]. Methods for balancing data from devices with different data should be implemented to ensure accurate detection of attack types and better overall IDS performance. The key strategy to solve this issue is the synthetic minority over-sampling technique. This method can generate synthetic samples for underrepresented classes and balance the data distribution without requiring more real-world data [94]. Furthermore, future research should enhance adaptive aggregation methods to address imbalances in federated IoT.

TABLE 10. Summary of challenges and future directions of FL in IoT.

Challenge	Results	Future research direction
Data heterogeneity	• Observed divergence in local model performance due to uneven data distribution among clients (nonIID data). Complicates the training process, as different devices contribute unequally to the global model, resulting in degrading model accuracy. • Difficult convergence when device data varies widely, especially with limited computational resources.	• Developing Research should be directed into developing algorithms that can dynamically select clients and cluster devices based on real-time data characteristics.
Privacy and Security	• Adoption of techniques such as differential privacy and secure aggregation to safeguard user data. Limited client participation in DP frameworks poses a critical challenge • Lower interaction may introduce distortions in aggregated models. • Increased computational overhead impacting performance.	• Enhance lightweight cryptographic protocols. • Optimize security measures to balance privacy with efficient computation and transmission. • Developing robust aggregation algorithms that are resilient to data poisoning attacks. Leveraging federated anomaly detection to identify malicious participants without central access to raw data. • Exploring dynamic resource allocation mechanisms to effectively balance the computational and storage requirements of integration Blockchain within the constraints of resource-limited federated learning.
Scalability	• Existing approaches struggle as IoT deployments scale, leading to communication overhead and performance bottlenecks. • Increased latency and resource allocation issues.	• Future studies should aim to enhance methods that minimize data transmission during the learning process, such as compression strategies, quantization, and gradient sparsification, to reduce the size of model updates • Edge-based aggregation to handle large-scale deployments. • Focus on system architecture design using multi-layered topologies in which edge devices coordinate local updates before global aggregation to reduce communication and computational load
Data imbalance	• Local datasets often suffer from skewed data distributions, causing model bias toward devices or classes with more data, reducing accuracy	• Integrate data re-balancing techniques and enhance adaptive aggregation methods to address imbalances in federated IoT. • Investigate synthetic data generation and resampling strategies for better representativeness.
Fairness	• Model bias arising from unequal distribution and participation among devices. • Inequalities in computational capabilities among devices may result in dominant contributions from more capable clients, excluding those with fewer resources, leading to discrepancies in performance across various user groups.	• Explore and develop fairness-aware aggregation algorithms for the distribution and allocation of resources to reduce communication challenges, decrease computational overhead, and enhance model accuracy.
Resource constraints	• IoT devices have limited computing, energy, and memory, resulting in restricted local model training.	• Optimize algorithms to reduce power consumption and improve computational efficiency. • Develop resource-adaptive training and communication mechanisms.
Communication Overhead	• High volumes of data interchange across devices and servers are prevalent, significantly impacting network bandwidth and energy consumption. • Frequent transmission rounds can cause latency issues.	• Future work should develop compression techniques to reduce communication overhead • Developing quantization solutions for minimizing the size of transmitted data.

E. ENSURING FAIRNESS

Addressing fairness issues in FL-based IDS is crucial for equal performance across various IoT devices, particularly in environments with diverse resources, data distributions, and participation levels. One of the strategies used is fair model aggregation to guarantee equal or weighted contributions to devices depending on variations in data amount, quality, or capabilities of devices [157]. Another strategy can be implemented using dynamic weights depending on device performance to guarantee fairness while retaining global model accuracy [158]. In addition, scheduling and prioritizing underrepresented participants during training results in more balanced contributions over time [159], [160]. Therefore, future research must focus on developing a fairness system for distributing and allocating resources, which is essential to reduce communication challenges, decrease computational overhead, and enhance model accuracy.

F. OPTIMIZING RESOURCE USAGE

Improving resource utilization in FL-based IDS systems is crucial for IoT contexts because devices often encounter energy, computation, and communication restrictions.

Communication efficiency can be optimized using approaches such as model compression, including sparsification, pruning, and quantization techniques, to decrease network load [85], [161], [162]. These solutions promote scalability and sustainability in IoT-based FIDS. This promotion ensures robust intrusion detection while conserving device performance. Optimizing resource usage in FL requires balancing efficiency, accuracy, privacy, and scalability. Future work should focus on adaptive and lightweight compression strategies and energy-efficient algorithms that are ideal for implementation on resource-constrained IoT devices.

G. IMPROVING COMMUNICATION OVERHEAD

Despite substantial research efforts, FL continues to face challenges related to efficient communication. The associated computational and storage requirements can significantly elevate as the number of edge nodes increases. Furthermore, transmitting local models to a central server is susceptible to delays or data loss due to fluctuations in network bandwidth. The heterogeneity in client dataset sizes further complicates timely and consistent model updates. Consequently, achieving an optimal trade-off between communication efficiency and model accuracy remains a critical research objective. Future work should develop techniques to reduce communication overhead and establish quantization solutions for minimizing the size of transmitted data.

VII. RESEARCH LIMITATIONS

Although the search strategy in this study was comprehensive, some pertinent publications may be missing owing to the exclusion of non-English studies, the selection of keywords, and database limitations.

VIII. CONCLUSION

This work conducts an SLR that examines recent studies in the domain of FL for IoT systems. The review underscores the crucial role of FL in enhancing privacy and efficiency within IoT environments. It also identifies critical gaps that must be addressed for the broader adoption of FL-based IDS solutions. Bridging these gaps will be essential to advancing future secure and resilient next-generation IoT infrastructures. In this study, 43 of 226 publications, which were published between 2020 and 2024, were analyzed and classified depending on their content. We first provided an overview of the latest developments in IoT, which offered a comprehensive understanding of this ecosystem over various important dimensions, including ML models, datasets, and techniques used in such works. We also discussed various open research challenges and future directions related to the FL-IoT architecture. The review reveals that integrating FL within IoT significantly enhances data privacy while reducing latency.

Regarding RQ1, we classified reviewed studies according to various characteristics, including their topical focus, digital library source, publication year, and publishing type. This overview provided insights into how FL research trends applied to IoT. RQ2 identified the datasets utilized to assess FL-based IDS models in IoT. The review found that common datasets such as NSL-KDD, CICIDS2019, and TON-IoT are predominantly used, which reveals a reliance on standard data sources. However, certain studies employ more diverse and updated datasets that capture the complexity of real-world IoT environments. RQ3 explored the ML techniques employed in developing IDSs to detect attacks on IoT devices, which provides an overview of current advancements in IoT security.

Recent studies have widely applied CNN, LSTM networks, and SVM, emphasizing FL to enhance privacy and increase the accuracy of detecting attacks in IoT environments.

This situation demonstrates ongoing innovation to enhance the efficiency and scalability of IDSs in IoT systems. RQ4 investigated the key challenges of adopting FL-based IDS approaches in IoT environments, which is crucial for understanding the obstacles to broader implementation. These issues highlight the complexity of deploying FL in IoT systems, which emphasizes the need for more efficient communication protocols and robust defense mechanisms to mitigate adversarial threats.

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